

$\Delta 133p53\alpha$ and $\Delta 160p53\alpha$ isoforms of the tumor suppressor protein p53 exert dominant-negative effect primarily by co-aggregation

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eLife Assessment

This **important** study investigates the molecular mechanisms by which the p53 isoforms $\Delta 133p53\alpha$ and $\Delta 160p53\alpha$ exert dominant-negative effects on full-length p53 (FLp53). Through a combination of chromatin immunoprecipitation, transcriptional reporter assays, subcellular localization analyses, and protein aggregation experiments, the authors provide **solid** evidence that these N-terminally truncated isoforms promote co-aggregation with FLp53, disrupting its transcriptional activity and cellular distribution. The revised manuscript successfully addresses prior reviewer concerns, and the findings are well supported by the experimental data.

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Abstract

p53 is a tumor suppressor protein with multiple isoforms with shared or specific functions. However, two of its isoforms, $\Delta 133p53\alpha$ and $\Delta 160p53\alpha$, with large N-terminal deletions, can cause cancer. These isoforms exert a dominant-negative effect on full-length p53 (FLp53), although the precise molecular mechanisms are unknown. Here, we investigate the mechanisms of action of $\Delta 133p53\alpha$ and $\Delta 160p53\alpha$ isoforms using chromatin immunoprecipitation, luciferase expression, subcellular fractionation, immunofluorescence assays, and apoptotic caspase activity assay. Our study elucidates that these DNA-binding deficient p53 isoforms form hetero-tetrameric complexes with FLp53 and disrupt FLp53's DNA binding and transcriptional activities when present in a higher proportion than FLp53 in the tetramer. However, these structurally unstable isoforms promote vigorous protein aggregation involving FLp53, disrupting its structure and sequestering it in the cytoplasmic and nuclear aggregates, thereby limiting its availability to function as a transcription activator protein. Thus, co-aggregation of $\Delta 133p53\alpha$ and $\Delta 160p53\alpha$ with FLp53, rather than hetero-tetramerization, is likely the primary factor contributing to their dominant-negative

effect. Modulating the stability and aggregation of p53 isoforms could be a novel strategy for cancer therapy.

Introduction

The tumor suppressor protein p53 is an essential transcription activator protein modulating the expression of multiple target genes. By regulating various cellular processes such as cell cycle arrest, apoptosis, senescence, DNA repair, and angiogenesis, p53 prevents cancer development^{1,2}. The intricate p53 pathway is stimulated by various stress stimuli, including DNA damage and oncogene activation, which lead to the stabilization and activation of the p53 protein³. Stabilized p53 acts as a transcription activator protein as it can directly bind to its response elements (REs) and recruit different transcription co-activator proteins at various gene promoters⁴. One of the best-known p53 targets is the p21 (also known as CIP1, WAF1) gene promoter, which contains two p53REs and is efficiently activated by the p53 protein^{5,6}. The encoded p21 protein blocks cell cycle progression and thereby executes tumor suppressor activity of the p53 protein^{7,8}.

The TP53 gene produces at least twelve distinct p53 isoforms: p53 (α , β , γ), $\Delta 40$ p53 (α , β , γ), $\Delta 133$ p53 (α , β , γ), and $\Delta 160$ p53 (α , β , γ). It has been shown that the p53 isoforms contribute to the regulation of the p53 pathway by regulating the expression of p53 target genes^{9,10,11}. Notably, p53 isoforms exhibit both overlapping and distinct functions compared to canonical p53^{11,12,13} and have been reported to promote tumor progression in various cancers^{11,14}. The p53 isoforms are generated through alternative promoter usage, alternative splicing, and alternative transcription starting site usage^{9,15}. These isoforms differ from the full-length (FL) p53 (canonical p53, p53 α) by lacking variable segments of the N-terminus (N-terminal isoforms: $\Delta 40$, $\Delta 133$, and $\Delta 160$) and possessing alternative C-terminus (C-terminal isoforms - α , β , and γ). The FLp53 protein consists of four primary functional domains: transactivation domains (TAD-1 and TAD-2), DNA-binding core domain (DBD), nuclear localization sequence (NLS), and the oligomerization domain (OD)¹⁶. Native FLp53 typically functions as a tetrameric protein, where the individual monomers bind cooperatively to DNA^{17,18}. The C-terminal truncated isoforms (β/γ -isoforms) lacking the OD have lower transactivation activity because of reduced tetramerization efficiency^{19,20,21,22}. In contrast, the N-terminal truncated isoforms (α -isoforms), which retain the entire C-terminal OD, can form various complexes with FLp53 and exert dominant-negative effects on p53's transcriptional activity^{23,24,25}. This study focuses on two large N-terminal deleted p53 isoforms: $\Delta 133$ p53 α and $\Delta 160$ p53 α , the latter has been largely unexplored. For convenience and simplicity, we have written $\Delta 133$ p53 and $\Delta 160$ p53 to represent the α isoforms ($\Delta 133$ p53 α and $\Delta 160$ p53 α) throughout this manuscript. The $\Delta 133$ p53 protein lacks both TADs and a part of the first conserved cysteine box of the DBD, whereas the $\Delta 160$ p53 protein lacks both TADs and the entire first conserved cysteine box of the DBD^{9,26}.

The $\Delta 133$ p53 isoform exhibit complex biological functions, with both oncogenic and non-oncogenic potentials. Recent studies demonstrate the non-oncogenic yet context-dependent role of the $\Delta 133$ p53 isoform in cancer development. $\Delta 133$ p53 expression has been reported to correlate with improved survival in patients with TP53 mutations^{27,28}, where it promotes cell survival in a non-oncogenic manner^{29,30}, especially under low oxidative stress³¹. Alternatively, other recent evidences emphasize the notable oncogenic functions of $\Delta 133$ p53 as it can inhibit p53-dependent apoptosis by directly interacting with the FLp53^{23,32}. The oncogenic function of the newly identified $\Delta 160$ p53 isoform is less known, although it is associated with p53 mutation-driven tumorigenesis³³ and in melanoma cells' aggressiveness³⁴. Whether or not the $\Delta 160$ p53 isoform also impedes FLp53 function in a similar way as $\Delta 133$ p53 is an open question. However, these p53 isoforms can certainly compromise p53-mediated tumor suppression by interfering with FLp53 binding to target genes such as p21 and miR-34a^{24,25} by dominant-negative effect, the exact mechanism is not known.

Protein aggregation has become a central focus of modern biology research and has documented implications in various diseases, including cancer³⁵, ³⁶, ³⁷. Protein aggregates can be of different types ranging from amorphous aggregates to highly structured amyloid or fibrillar aggregates, each with different physiological implications. In the case of p53, whether protein aggregation, and in particular, co-aggregation with large N-terminal deletion isoforms, plays a mechanistic role in its inactivation is yet underexplored. Interestingly, the $\Delta 133p53\beta$ isoform has been shown to aggregate in several human cancer cell lines³⁸. Additionally, the $\Delta 40p53\alpha$ isoform exhibits a high aggregation tendency in endometrial cancer cells³⁹. Although no direct evidence exists for $\Delta 160p53$ yet, these findings imply that p53 isoform aggregation may play a major role in their mechanisms of actions.

The precise molecular mechanisms involved in the biological function of the $\Delta 133p53$ and the $\Delta 160p53$ proteins, especially how they exert dominant-negative effects on FLp53, remain poorly understood. In this study, we have analyzed the DNA-binding, transcriptional activity, and oligomerization capacity of the human FLp53, $\Delta 133p53$, and $\Delta 160p53$ proteins. Our data show that the $\Delta 133p53$ and $\Delta 160p53$ proteins are deficient in binding to p53-regulated promoters and inducing expression of the p53 target genes. Moreover, the $\Delta 133p53$ and $\Delta 160p53$ isoforms can form hetero-oligomers with FLp53 and inhibit FLp53 function in a concentration-dependent manner. Two potential mechanisms for their dominant-negative effects have been suggested: (i) the formation of hetero-tetramers with FLp53, inhibiting its DNA-binding and transcriptional activity, and (ii) the aggregation property of the isoforms leading to co-aggregation of the FLp53 protein. Interestingly, while hetero-tetramerization partially attenuates p53 activity, the co-aggregation of FLp53 with $\Delta 133p53$ and $\Delta 160p53$ emerges as the predominant factor contributing to the dominant-negative effect.

Results

$\Delta 133p53$ and $\Delta 160p53$ proteins exhibit structural destabilization, exposure of hydrophobic surfaces, and high propensity to aggregation

The DBD, also known as the core domain of the FLp53 protein (human p53, residues 94-292) has a structure with an aggregation-prone sequence (residues 251–257)⁴⁰ (Figure 1A). The core domain contains a hydrophobic core of β -sandwich, which consists of two antiparallel β -sheets with four strands (S1, S3, S8, and S5) and five strands (S6, S7, S4, S9, and S10) respectively⁴¹ (Supplementary Figure 1A). The DNA-binding surface is constituted by two large loops (L2 and L3) and a loop-sheet-helix (LSH) motif, which comprises loop L1, a β -sheet formed by the β -hairpin S2–S2', the C-terminal residues of strand S10, and the α -helix H2⁴² (Supplementary Figure 1A). Compared to FLp53, the N-terminal deletion isoforms $\Delta 133p53$ and $\Delta 160p53$ lack the entire TADs (Figure 1A), which are crucial for recruiting transcription co-activator proteins⁴³. In addition, the $\Delta 133p53$ protein is missing the β -strand S1 in the β -sandwich and a part of the LSH motif (L1, S2, and the N-terminal residue 132 of the strand S2') in the DBD (Supplementary Figure 1A). The $\Delta 160p53$ protein, on the other hand, lacks a part of the β -sandwich (β -strands S1, S3, and N-terminal residues 156-159 of S4) and a part of the LSH motif (L1 and β -hairpin S2–S2') (Supplementary Figure 1A). Therefore, the N-terminal deletion affects the structural integrity of both the β -sandwich and the LSH motif in the core domain.

To assess the structural impact of the N-terminal deletions in FLp53, we analyzed the contact information derived from the core domain of FLp53 (PDB ID: 3KMD) structure⁴⁴. We observed that both $\Delta 133p53$ and $\Delta 160p53$ lack several crucial contacts between the N- and C-terminal regions in FLp53 (Figure 1B). For $\Delta 133p53$, the contact between the L1 and S2 segments in the N-terminal region and the α -helix H2 in the C-terminal region within the LSH motif is disrupted

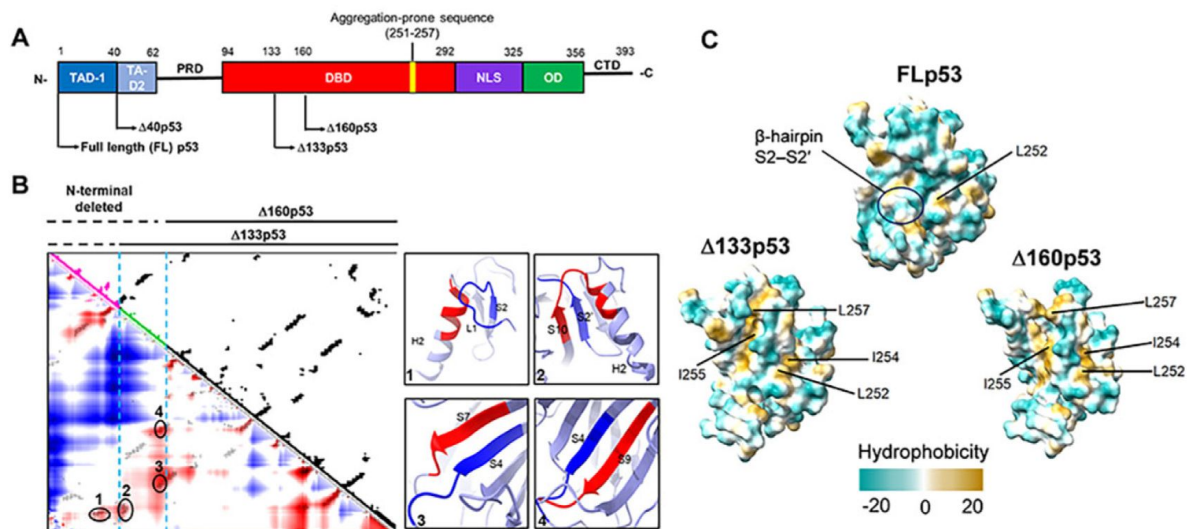


Figure 1.

The $\Delta 133$ p53 and $\Delta 160$ p53 proteins exhibit structural destabilization.

(A) Schematic domain structures of full-length (FL) p53 and its isoforms. TAD, transactivation domain; PRD, proline-rich domain; DBD, DNA-binding domain; NLS, nuclear localization sequence; OD, oligomerization domain; CTD, C-terminal domain. $\Delta 133$ p53 lacks the entire TAD and approximately 20% of the DBD. $\Delta 160$ p53 lacks the entire TAD and approximately 33% of the DBD. An aggregation region ILTIITL (residues 251-257) is indicated in the DBD domain. **(B)** Contact density map (C- α 8Å) of the core domain of FLp53 (PDB ID: 3KMD) using CWView⁸⁸. The residues 94-132 and 133-159 were colored magenta and green, respectively. Higher densities are shown in red, and low densities are shown in blue. Among these contacts, four higher-density contacts were selected and shown in oval circles on the map. The corresponding secondary structures of the contacts between the deleted region (blue) and the region (red) in the loop-sheet-helix (LSH) motif or the β -sandwich are shown on the right. **(C)** Comparison analysis of surface hydrophobicity of the core domain in FLp53 (PDB ID: 3KMD) and its isoforms $\Delta 133$ p53 and $\Delta 160$ p53 by ChimeraX^{89, 90}. The exposed hydrophobic residues in the aggregation region ILTIITL (residues 251-257) are labeled. The color scale of hydrophobicity is indicated with the maximum and minimum values in dark cyan and dark goldenrod, respectively. The β -hairpin S2-S2' was shown in an oval circle.

(**Figure 1B** [↗](#), insert 1), suggesting a potential impairment of the stabilizing of DNA-binding interface. For $\Delta 160p53$, in addition to the missing contact between L1, S2 and H2, there are also no contacts between the S2' segment in the N-terminal region and the S10 and H2 segments in the C-terminal region within the LSH motif (**Figure 1B** [↗](#), insert 2), as well as between S4 and S7 or S9 in the β -sandwich core domain (**Figure 1B** [↗](#), inserts 3 and 4). These contacts are crucial for maintaining the structural integrity and folding of the core domain. Therefore, both $\Delta 133p53$ and $\Delta 160p53$ will likely have a partially or fully destabilized core domain that may impair their DNA-binding ability.

We predicted the monomer structure of $\Delta 133p53$ and $\Delta 160p53$ by AlphaFold2⁴⁵ [↗](#) (Supplementary Figure 1B). Hydrophobic surface analysis revealed that $\Delta 133p53$ and $\Delta 160p53$ exposed four highly hydrophobic residues (L252, I254, I255, and L257) within the aggregation-prone sequence, while FLp53 exposes only one residue (L252) (**Figure 1C** [↗](#)). This increased hydrophobicity may induce the aggregation of $\Delta 133p53$ and $\Delta 160p53$. Moreover, $\Delta 133p53$ and $\Delta 160p53$ have a partial or complete structural loss of the β -hairpin S2–S2' (Supplementary Figure 1A), which restricts access to the hydrophobic core⁴⁶ [↗](#). This structural alteration results in an enhanced exposure of the hydrophobic core of the β -sandwich domain compared to FLp53 (**Figure 1C** [↗](#)). This exposure potentially facilitates the propensity for protein aggregation, as suggested by previous studies⁴⁶ [↗](#), ⁴⁷ [↗](#) and our *in vitro* aggregation assay with purified FLp53, $\Delta 133p53$ and $\Delta 160p53$ proteins (Supplementary Figure 2). In summary, these findings suggest that the large N-terminal deletion in $\Delta 133p53$ and $\Delta 160p53$ would compromise the structural stability of FLp53 and induce aggregation, with $\Delta 160p53$ being more severely affected among the two.

$\Delta 133p53$ and $\Delta 160p53$ form complexes with FLp53

To investigate the oligomerization properties of $\Delta 133p53$ and $\Delta 160p53$, which retain the OD known to facilitate p53 tetramerization⁴⁸ [↗](#), we co-expressed the FLAG-tagged p53 proteins as the baits along with the V5-tagged FLp53, $\Delta 133p53$, and $\Delta 160p53$ proteins as preys. The V5-tagged FLp53, $\Delta 133p53$, and $\Delta 160p53$ proteins specifically interacted with their corresponding FLAG-tagged proteins (**Figure 2A** [↗](#), lanes 1–6), demonstrating their capacity to form homo-oligomers (**Figure 2B** [↗](#)). In addition, interactions between FLp53-FLAG/ $\Delta 133p53$ -V5, FLp53-FLAG/ $\Delta 160p53$ -V5, $\Delta 133p53$ -FLAG/ $\Delta 160p53$ -V5, and $\Delta 160p53$ -FLAG/ $\Delta 133p53$ -V5 were found (**Figure 2A** [↗](#), lanes 7–10), suggesting that they can also form hetero-oligomers (**Figure 2B** [↗](#)). These findings are consistent with previous studies indicating that $\Delta 40p53$ can hetero-oligomerize with FLp53⁴⁹ [↗](#), ⁵⁰ [↗](#). To rule out the possibility that the observed interactions between FLp53 and its isoforms $\Delta 133p53$ and $\Delta 160p53$ were artifacts caused by the FLAG and V5 antibody epitope tags, we co-expressed FLAG-tagged FLp53 with untagged $\Delta 133p53$ and $\Delta 160p53$. Immunoprecipitation assays demonstrated that FLAG-tagged FLp53 could still interact with the untagged $\Delta 133p53$ and $\Delta 160p53$ isoforms (Supplementary Figure 3, lanes 3 and 4), confirming the formation of hetero-oligomers between FLp53 and its isoforms. These findings demonstrate that $\Delta 133p53$ and $\Delta 160p53$ can oligomerize with FLp53 and with each other.

$\Delta 133p53$ and $\Delta 160p53$ negatively influence FLp53's DNA-binding activity

Previous studies have shown that the p53 lacking the TADs and the PRD ($\Delta Np53DBD$, residues 94–393) retain full DNA-binding capacity as the FLp53⁵¹ [↗](#), ⁵² [↗](#). In contrast, the $\Delta 133p53$ (residues 133–393) and $\Delta 160p53$ (residues 160–393) isoforms additionally lack part of the DBD (residues 94–292). This deficiency raises questions about the impact of partial DBD deletion on the DNA-binding activity of p53. To address this question, we performed a chromatin immunoprecipitation (ChIP) assay to measure the DNA-binding activity of FLp53 and its isoforms $\Delta 133p53$ and $\Delta 160p53$ to several known p53 targeted promoters controlling proteins regulating cycle arrest (p21), protein stability (MDM2), and apoptosis induction (BAX and PUMA) (**Figure 3A** [↗](#)). Previous studies have classified p53 DNA-binding affinities. Accordingly, the p21 promoter has the highest affinity, the

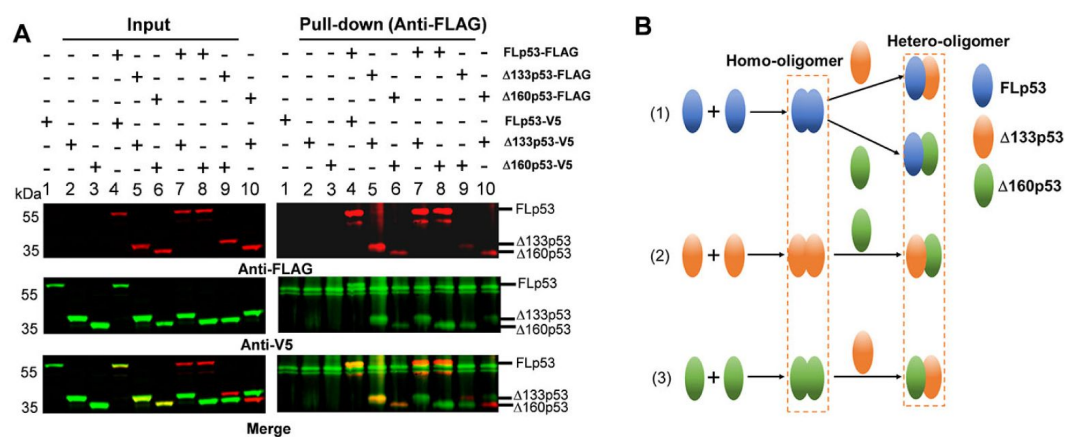


Figure 2.

Complex formation among FLp53 and its isoforms Δ133p53 and Δ160p53.

(A) Co-immunoprecipitation of FLp53-FLAG, Δ133p53-FLAG, and Δ160p53-FLAG along with the FLp53-V5, Δ133p53-V5, and Δ160p53-V5 proteins in H1299 cells. The complexes were isolated with anti-FLAG antibody-coupled magnetic beads. Western blot analysis was done with the anti-FLAG and anti-V5 antibodies. The merged image corresponds to the overlay between the FLAG and V5 antibody signals. Input corresponds to 5% of the whole cell lysate. **(B)** A summary of FLp53, Δ133p53, and Δ160p53 oligomerization based on data shown in panel A.

MDM2 promoter has moderate affinity, whereas the BAX and PUMA promoters show relatively low p53-binding affinity^{53, 54, 55}. Consistent with these findings, our results revealed distinct binding affinities of the FLp53 protein to these target gene promoters (**Figure 3B**). Notably, the $\Delta 133p53$ and $\Delta 160p53$ isoforms demonstrated a significant loss of DNA-binding activity at these promoters (**Figure 3B**).

We subsequently hypothesized that $\Delta 133p53$ and $\Delta 160p53$ might compromise the DNA-binding ability of FLp53 through hetero-oligomer formation (**Figure 2**). To study this, we co-transfected FLp53 along with either $\Delta 133p53$ or $\Delta 160p53$ expressing plasmid and measured FLp53's binding to the aforementioned promoters. The presence of $\Delta 133p53$ or $\Delta 160p53$ significantly reduced the binding of FLp53 to the p21 promoter, which has the highest affinity for FLp53, while having minimal impact on its interaction with the other promoters (**Figure 3C**). These findings suggest that $\Delta 133p53$ and $\Delta 160p53$ isoforms may selectively impair the ability of FLp53 to bind to high-affinity target promoters.

$\Delta 133p53$ and $\Delta 160p53$ impede FLp53-mediated transcription activation

The p53-mediated transcription activation depends on its binding ability to specific p53REs at the target gene promoters^{56, 57}. To evaluate the influence of the FLp53 protein interaction with its isoforms, $\Delta 133p53$ and $\Delta 160p53$, on transcription, we employed a luciferase reporter assay with the p21, MDM2, BAX, and PUMA promoters fused to a firefly luciferase reporter gene. As expected, all four promoters were responsive to FLp53-mediated transcriptional activation (**Figure 4A-D**). Notably, FLp53 activation of the PUMA promoter was relatively low, possibly due to the lower binding affinity of FLp53 to the PUMA promoter (**Figure 3B**). In contrast, neither $\Delta 133p53$ nor $\Delta 160p53$ activated any tested promoters (**Figure 4A-D**), indicating a lack of functional capability to stimulate transcription from these promoters.

In order to assess how $\Delta 133p53$ and $\Delta 160p53$ influence FLp53-mediated transcription on different p53-responsive promoters, we co-expressed FLp53 along with increasing amounts of $\Delta 133p53$ or $\Delta 160p53$ in H1299 cells. We found that co-expression of FLp53 with $\Delta 133p53$ or $\Delta 160p53$ at a 1:1 ratio did not significantly alter the activation of the p21 and MDM2 promoters compared to FLp53 alone (**Figure 4A** and **B**). This result suggests that at an equal protein expression ratio (**Figure 2A**), these isoforms did not exert a dominant-negative effect on the FLp53 at these promoters. However, a strong reduction in BAX promoter activation was evident under the same experimental condition (**Figure 4C**), suggesting that the isoforms can disrupt FLp53 function on the BAX promoter. Interestingly, both isoforms enhanced the activation of the PUMA promoter when added with FLp53 in a 1:1 ratio (**Figure 4D**).

The elevated $\Delta 133p53$ protein modulates p53 target genes such as miR-34a and p21, facilitating cancer development^{24, 58}. To mimic conditions where isoforms are upregulated relative to FLp53, we increased the ratios to 1:5 and 1:10. Adjusting the ratio of FLp53 to isoforms to 1:5 revealed that higher isoform expression levels significantly impaired FLp53-mediated activation of the p21, MDM2, and BAX promoters (**Figure 4A-C**). Interestingly, $\Delta 133p53$ at a 1:5 ratio had almost no impact on PUMA promoter activity, whereas $\Delta 160p53$ reduced it compared to FLp53 alone (**Figure 4D**). These results indicate that higher expression ratios of $\Delta 133p53$ and $\Delta 160p53$ relative to FLp53 might be required to exert dominant-negative effects. To test this hypothesis, we escalated the ratio of FLp53 to isoforms to 1:10. As expected, the activity of all four promoters decreased significantly at this ratio (**Figure 4A-D**). Notably, $\Delta 160p53$ showed a more potent inhibitory effect than $\Delta 133p53$ at the 1:5 ratio on all promoters except for the p21 promoter, where their impacts were similar (**Figure 4E-H**). However, at the 1:10 ratio, $\Delta 133p53$ and $\Delta 160p53$ had similar effects on all transactivation except for the MDM2 promoter (**Figure 4E-H**). This differential impact underscores the variability in how p53 isoforms influence FLp53 function across different promoters and expression ratios. Collectively, these findings demonstrate that

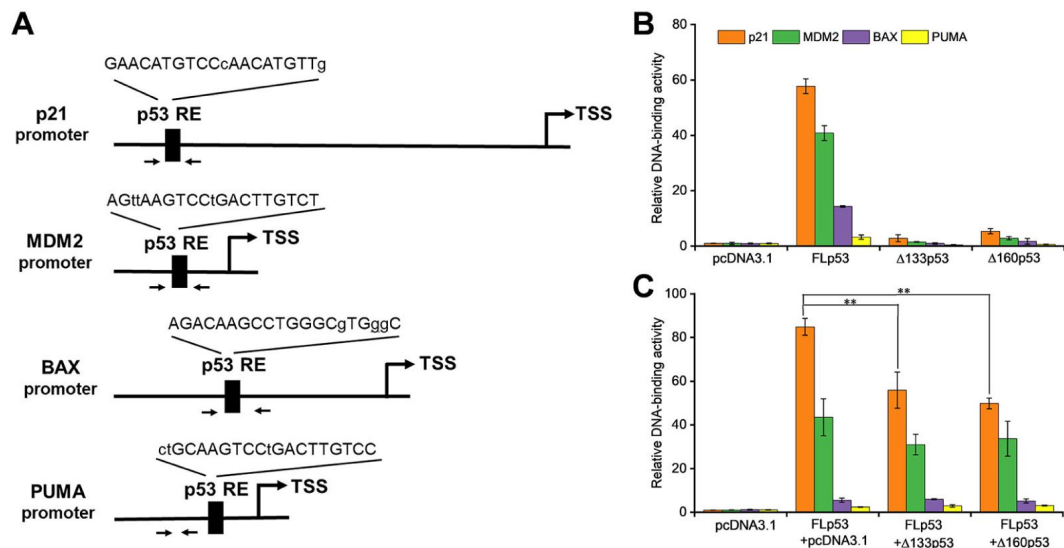


Figure 3.

Δ133p53 and Δ160p53 negatively influence FLp53's DNA-binding activity.

(A) Schematic representation of the promoters of p53 target genes (p21, MDM2, BAX, and PUMA). An arrow denotes the transcription start site (TSS). The DNA sequences of p53 response elements (REs) are shown, with uppercase and lowercase letters indicating matched and mismatched bases, respectively, in relation to the canonical p53 RE sequence (RRRCWWGYYY). Paired arrows highlight the regions subjected to quantitative PCR (qPCR) amplification. Specifically, the distal (5') p53 RE within the p21 gene promoter was analyzed. **(B)** Relative DNA-binding of the FLp53-FLAG, Δ133p53-FLAG, and Δ160p53-FLAG proteins to p53 target genes (p21, MDM2, BAX, and PUMA) in H1299 cells. **(C)** Relative DNA-binding of the FLp53-FLAG protein to the p53-target gene promoters in the presence of the V5-tagged protein Δ133p53 or Δ160p53 at a 1: 1 ratio. ChIP-qPCR assay data are shown as relative enrichment of promoter sequences of the target genes after normalization to the control, pcDNA3.1 plasmid transfected cells. Data are represented as the mean of technical triplicates $\pm$ standard deviation (SD), $**P < 0.01$ (Student's t-test).

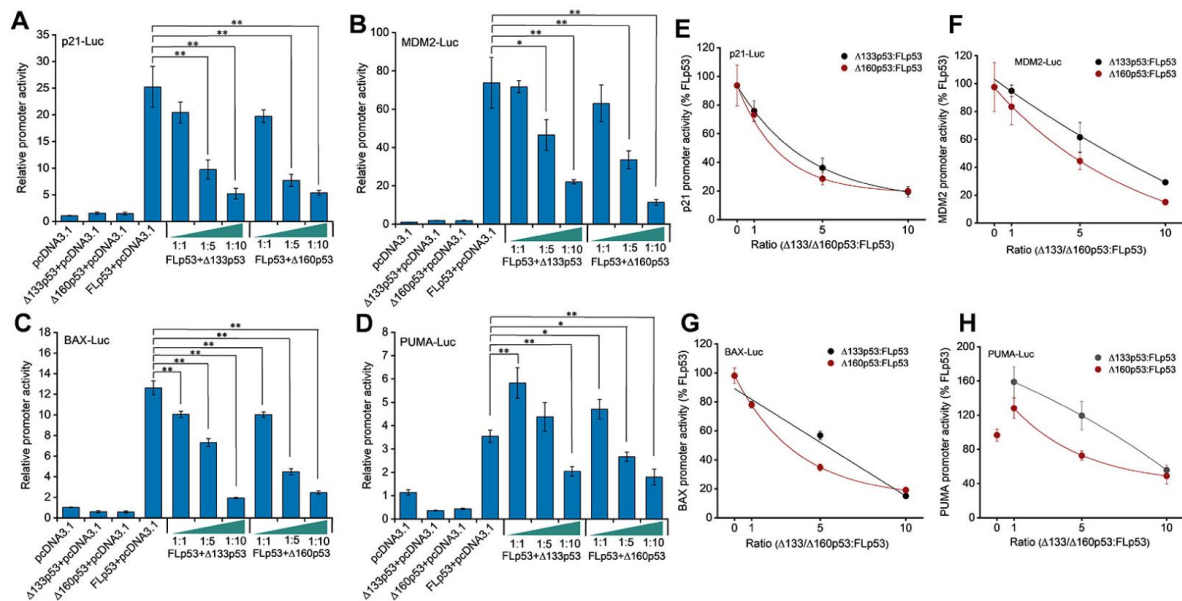


Figure 4.

Impact of Δ133p53 and Δ160p53 on FLp53-activated transactivation.

(A-D) H1299 cells were transfected with luciferase reporter plasmids driven by p21 (A), MDM2 (B), BAX (C), and PUMA (D) promoters, along with vectors expressing FLp53, Δ133p53, or Δ160p53. To evaluate the influence of the isoforms on FLp53's transactivation capability, co-expression was performed at ratios of 1:1, 1:5, and 1:10 relative to FLp53. Basal promoter activity was determined by transfecting cells with the empty vector, pcDNA3.1. Relative promoter activity is shown after normalization to the pcDNA3.1-treated sample activity. Data represent mean values $\pm$ standard deviation (SD) ($n=3$). $*P < 0.05$; $**P < 0.01$ (Student's t -test). (E-H) The transcriptional activity of FLp53 on the p21 (E), MDM2 (F), BAX (G), and PUMA (H) promoters was inhibited by Δ133p53 and Δ160p53. Inhibition curve fitting was performed using the exponential function ExpDec1 in Origin 2018 software.

isoforms with impaired DNA-binding capability can indeed exert dominant-negative effects on the function of FLp53, with the magnitude of this impact being promoter-specific and dependent on the relative expression levels of the proteins.

The $\Delta 133p53$ and $\Delta 160p53$ proteins affect the subcellular localization and aggregation of FLp53

p53 exerts its transcriptional activity in the nucleus, and its appropriate subcellular localization is crucial for regulating p53 functions⁵⁹. Immunoblotting analysis with isolated nuclear and cytoplasmic fractions revealed distinct subcellular localization patterns for the FLp53 and its isoforms $\Delta 133p53$ and $\Delta 160p53$ (**Figure 5A** and **B**). The cytoplasm-to-nucleus ratio of $\Delta 133p53$ and $\Delta 160p53$ was approximately 1.5-fold higher than that of FLp53 (**Figure 5B**). These results suggest that $\Delta 133p53$ and $\Delta 160p53$ exhibit a greater tendency for cytoplasmic localization.

Immunofluorescence analysis of the cells expressing FLp53 and the two isoforms confirmed the above-observed protein localization pattern (**Figure 5C**). The FLp53 protein was predominantly localized in the nucleus, with minimal to no fluorescence signal detected in the cytoplasm (**Figure 5C**, panel 1 and Supplementary Figure 4, panels 1 and 2). In contrast, the $\Delta 133p53$ and $\Delta 160p53$ proteins exhibited reduced nuclear staining and enhanced cytoplasmic staining relative to FLp53, with $\Delta 160p53$ showing more pronounced cytoplasmic staining (**Figure 5C**). Notably, $\Delta 133p53$ and $\Delta 160p53$ demonstrated a heterogeneous distribution in the nucleus and cytoplasm (**Figure 5C**, panels 2 and 3). Furthermore, both $\Delta 133p53$ and $\Delta 160p53$ exhibited predominantly cytoplasmic punctate staining, particularly $\Delta 160p53$, suggesting the formation of protein aggregates (**Figure 5C**, panels 2 and 3).

Given that p53 aggregation has been found in numerous tumor cells^{60, 61, 62, 63}, it is important to explore the effects of $\Delta 133p53$ or $\Delta 160p53$ isoforms on the aggregation of FLp53. To investigate the aggregation property of the p53 isoforms, we performed a subcellular fractionation of the H1299 cells expressing FLp53 along with $\Delta 133p53$ or $\Delta 160p53$ at 1:5 ratio. Notably, our fractionation approach allows us to analyze soluble cytoplasmic and nuclear fractions as well as insoluble nuclear fraction. Immunoblotting of the soluble fraction showed that an excess of $\Delta 133p53$ and $\Delta 160p53$ isoforms reduced the levels of FLp53 in both the cytoplasm and nucleus, with a notable reduction in the nucleus (**Figure 6A** and **C**). Conversely, insoluble nuclear FLp53 increased significantly upon co-expression with the p53 isoforms, with $\Delta 160p53$ showing a more prominent effect (**Figure 6B** and **D**, Supplementary Figure 5).

To exclude the possibility that FLAG or V5 tags contribute to protein aggregation, we also conducted subcellular fractionation of H1299 cells expressing FLAG-tagged FLp53 along with untagged $\Delta 133p53$ or $\Delta 160p53$ at a 1:5 ratio. The results showed (Supplementary Figure 6) a similar distribution of FLp53 across cytoplasmic, nuclear, and insoluble nuclear fractions as in the case of tagged $\Delta 133p53$ or $\Delta 160p53$ (**Figure 6A** to **D**). Notably, the aggregation of untagged $\Delta 133p53$ or $\Delta 160p53$ markedly promoted the aggregation of FLAG-tagged FLp53 (Supplementary Figure 6B and D), demonstrating that the antibody epitope tags themselves do not contribute to protein aggregation. These results collectively demonstrate that $\Delta 133p53$ and $\Delta 160p53$ isoforms strongly induce FLp53 aggregation, and the decrease in soluble FLp53 levels could be attributed to its aggregation when co-expressed with isoforms $\Delta 133p53$ and $\Delta 160p53$.

We complemented the immunoblotting assays with immunofluorescence assays. Our results clearly show the co-localization of FLp53 with the isoforms $\Delta 133p53$ and $\Delta 160p53$. More importantly, we find co-aggregation of FLp53 with the two isoforms forming punctate spots in the cytoplasm and nucleus (**Figure 6E**, panels 2 and 3 and Supplementary Figure 4, panels 3 and 4). Furthermore, more prominent subnuclear foci of FLp53 formed in the cells co-expressing the $\Delta 160p53$ protein (**Figure 6E**, panel 3 and Supplementary Figure 4, panel 4), which were detected

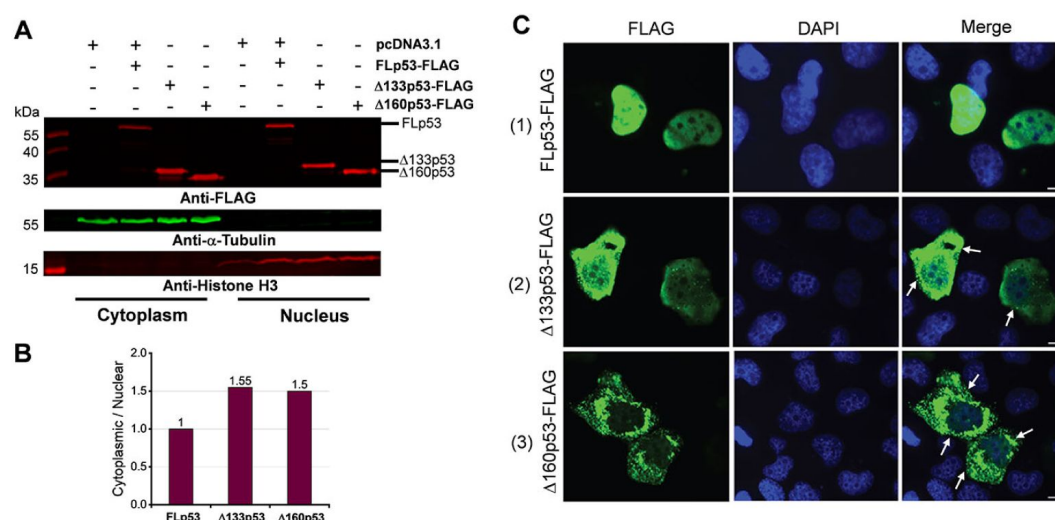


Figure 5.

Subcellular distribution of the FLP53, Δ133p53 and Δ160p53 proteins.

(A) Western blot analysis of subcellular fractionation of H1299 cells transfected with the FLAG-tagged FLP53, Δ133p53, or Δ160p53 proteins. Representative immunoblots indicate the presence of these proteins in different cellular fractions. α-Tubulin (cytoplasmic marker) and Histone H3 (nuclear marker) were used to ensure fraction integrity. **(B)** Quantitative bar graph summarizing the cytoplasm-to-nucleus ratio of protein levels. FLAG-tagged protein levels were normalized to their respective fractionation markers: α-Tubulin for the cytoplasmic fraction and histone H3 for the nuclear fraction. The relative ratios of Δ133p53 and Δ160p53 protein levels are displayed, with the FLP53 cytoplasm-to-nucleus ratio set as 1. **(C)** Immunofluorescence analysis of H1299 cells expressing FLAG-tagged FLP53, Δ133p53 and Δ160p53. Cell nuclei are visualized with DAPI (blue). The punctate spots (white arrows) indicating protein aggregates are marked. Scale bar, 50 μm.

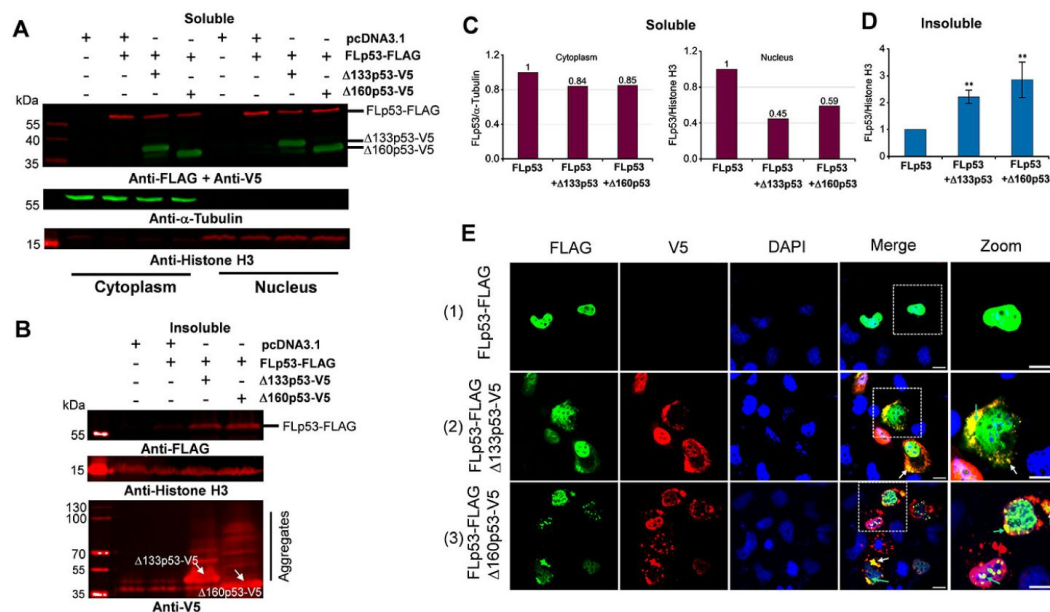


Figure 6.

Induction of FLP53 aggregation by Δ133p53 and Δ160p53 isoforms.

(A-B) Western blot analysis of soluble cytoplasmic and nuclear subcellular fractions **(A)** and insoluble nuclear fraction **(B)**. Biochemical fractionation of H1299 cells transfected with the FLAG-tagged FLP53 and V5-tagged Δ133p53 or Δ160p53 at a 1:5 ratio. The aggregated monomers of V5-tagged Δ133p53 and Δ160p53 (indicated by white arrows), as well as the higher molecular weight aggregates, are shown. This figure shows one representative experiment; $n=3$. **(C)** Quantitative bar graph summarizing the levels of the soluble FLP53-FLAG protein in the cytoplasmic and nuclear fractions. The FLP53-FLAG protein levels were normalized to the corresponding fractionation marker (α-Tubulin or histone H3). Relative accumulation of the FLP53-FLAG in Δ133p53-V5 or Δ160p53-V5 expressing cells is shown after considering the FLP53-FLAG protein expressing sample as 1. **(D)** Quantitative bar graph summarizing the levels of the FLP53-FLAG protein in the insoluble nuclear fraction relative to the histone H3. Data normalization as in panel C, data represent mean values $\pm$ standard deviation (SD). $^{**}P < 0.01$ (Student's t-test). **(E)** Immunofluorescence analysis of H1299 cells transfected with FLP53-FLAG alone or in combination with V5-tagged isoforms Δ133p53 and Δ160p53 expressing plasmids at a concentration ratio of 1:5. Cell nuclei were visualized with DAPI (blue). The co-aggregation of isoforms with FLP53 in the cytoplasm (white arrows) and nucleus (bright green arrows) are indicated. Scale bar, 50 μ m.

in the $\Delta 133p53$ protein expressing H1299 cells. Thus, these results imply that $\Delta 133p53$ and $\Delta 160p53$ isoforms efficiently induce co-aggregation of FLp53, potentially influencing its function in cancer cells.

The $\Delta 133p53$ and $\Delta 160p53$ proteins block pro-apoptotic function of FLp53

One of the physiological read-outs of FLp53 is its ability to induce apoptotic cell death⁶⁴. To investigate the effects of p53 isoforms $\Delta 133p53$ and $\Delta 160p53$ on FLp53-induced apoptosis, we measured caspase-3 and -7 activities in H1299 cells expressing different p53 isoforms (**Figure 7**). Caspase activation is a key biochemical event in apoptosis, with the activation of effector caspases (caspase-3 and -7) ultimately leading to apoptosis⁶⁵. The caspase-3 and -7 activities induced by FLp53 expression was approximately 2.5 times higher than that of the control vector (**Figure 7**). Co-expression of FLp53 and the isoforms $\Delta 133p53$ or $\Delta 160p53$ at a ratio of 1: 5 significantly diminished the apoptotic activity of FLp53 (**Figure 7**). This result aligns well with our reporter gene assay, which demonstrated that elevated expression of $\Delta 133p53$ and $\Delta 160p53$ impaired the expression of apoptosis-inducing genes BAX and PUMA (**Figure 4G** and **H**). Moreover, a reduction in the apoptotic activity of FLp53 was observed irrespective of whether $\Delta 133p53$ or $\Delta 160p53$ protein was expressed with or without a FLAG tag (**Figure 7**). This result, therefore, also suggests that the FLAG tag does not affect the apoptotic activity or other physiological functions of FLp53 and its isoforms. Overall, the overexpression of p53 isoforms $\Delta 133p53$ and $\Delta 160p53$ significantly attenuates FLp53-induced apoptosis, independent of the protein tagging with the FLAG antibody epitope.

Discussion

The inactivation of p53 is one of the most common events in cancer cells⁶⁶. Although functionally compromised p53 mutations are prevalent in more than half of human cancers^{67, 68}, specific p53 isoforms have recently been associated with tumorigenesis in various cancer types^{11, 14}. The aberrant expression of $\Delta 133p53$ has been found in colorectal cancer^{24, 69} and lung cancer⁵⁸. In comparison, the oncogenic role of the $\Delta 160p53$ isoform in human cancers is less explored¹¹, although links have been established between the expression of $\Delta 160p53$ isoform and melanoma cells³⁴. The mechanisms of the oncogenic potential of p53 isoforms have attracted attention from researchers^{14, 38, 39, 70, 71, 72}. A theory of the dominant-negative effect has been introduced, but its exact details have remained unknown.

The ChIP and luciferase activity assays show that the isoforms $\Delta 133p53$ and $\Delta 160p53$ are defective in their ability to bind DNA and activate transcription (**Figure 3B** and **Figure 4A-D**). A possible reason for the diminished DNA-binding function of these isoforms is that their core domains are defective (**Figure 1B**). The $\Delta 133p53$ and $\Delta 160p53$ isoforms lack approximately 20% and 33% of the DBD, which hinders their ability to bind to p53REs. We find that co-expression of $\Delta 133p53$ and $\Delta 160p53$ with FLp53 exerts an inhibitory effect on FLp53's DNA-binding ability (**Figure 3C**). However, the degree to which $\Delta 133p53$ and $\Delta 160p53$ impair FLp53 function is influenced by both the expression levels of these isoforms and the specific promoters of the target genes (**Figure 4**). For the p21 and MDM2 promoters, equal expression of FLp53 and the two isoforms didn't show much effect. However, higher expression levels of the isoforms gradually inhibited FLp53's activity to induce transcription from those promoters (**Figure 4A** and **B**). Notably, the PUMA promoter behaved differently, whereby 1:1 co-expression of FLp53 with $\Delta 133p53$ or $\Delta 160p53$ significantly increased its activity (**Figure 4D**). We speculate that this isoform expression level may activate p21 induction pathways independent of FLp53⁷³, potentially facilitating the interaction between FLp53 and p21, which is required for optimal

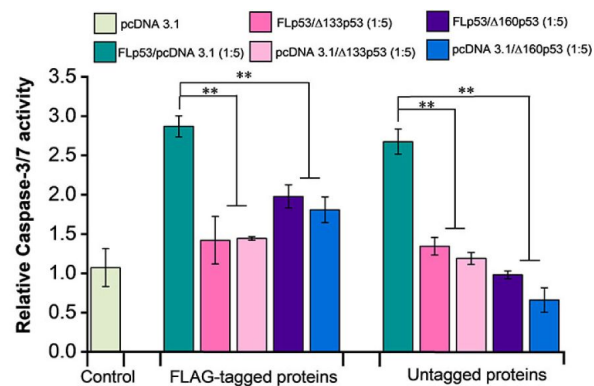


Figure 7.

Overexpression of $\Delta 133p53$ and $\Delta 160p53$ block pro-apoptotic activity of FLp53.

Caspase-3 and -7 activities were detected in H1299 cells co-transfected with plasmid expressing FLp53 or the isoforms $\Delta 133p53$ and $\Delta 160p53$ at a 1:5 ratio. Experiments were performed with the plasmids expressing FLAG-tagged and untagged FLp53, $\Delta 133p53$ and $\Delta 160p53$. The relative activities of caspase-3 and -7 were normalized to those of the control vector (transfected only with the pcDNA3.1 plasmid). Data represent mean values $\pm$ standard deviation (SD). $^{***}P < 0.01$ (Student's t-test).

activation of PUMA promoter activity⁷⁴. These results underscore a complex regulatory network where FLp53 and the isoform expression levels, target gene specificity, and transcriptional activities modulate p53-mediated cellular responses.

Our co-expression data clearly indicate that the isoforms $\Delta 133p53$ and $\Delta 160p53$ exert a dominant-negative effect on FLp53 function (**Figures 3**, **4** and **7**). Since both of these p53 isoforms are defective in DNA binding and they possess an intact C-terminal OD, it is not unexpected that they form different species of homo- and hetero-tetramers with FLp53. In accordance with that, the immunoprecipitation assays using FLAG-tagged FLp53 and V5-tagged/untagged $\Delta 133p53$ or $\Delta 160p53$ demonstrate a potential direct interaction between these isoforms and FLp53 (**Figure 2A**, lanes 7 and 8 and Supplementary Figure 3, lanes 3 and 4). However, in what ratio the p53 isoforms impose inhibition to FLp53 function is worth investigating. We employed a theoretical model of p53 hetero-tetramer activity in transcriptional activation from p21, MDM2, BAX, and PUMA promoters by varying relative expression levels of the isoforms⁷⁵ (Supplementary Figure 7). We assumed that the transfection level was a good proxy for the expression level. Interestingly, the experimental inhibition curves with $\Delta 133p53$ or $\Delta 160p53$ derived from the luciferase reporter assay were all above the theoretical inhibition curves for two isoform molecules per tetramer (Supplementary Figure 7B-E), suggesting that at least three isoform molecules per tetramer are required to abolish the transcriptional activity of FLp53. This is similar to an earlier report with $\Delta 40p53$ isoform, which was shown to exert a dominant-negative effect at a 3:1 ratio with FLp53⁵⁰.

Interestingly, similar to $\Delta 40p53$ ³⁹, the $\Delta 133p53$, and $\Delta 160p53$ proteins can aggregate in the cytoplasm (**Figures 5C** and **6E**). Due to cytoplasmic aggregation and a stronger tendency for cytoplasmic localization (**Figure 5C**), only a limited amount of these isoforms can successfully be imported into the nucleus successfully. Additionally, the isoforms tend to aggregate within the nucleus (**Figure 6B** and **E**). Consequently, the ratio of these isoforms to FLp53 diminishes in the nucleus in contrast to their expression levels. This decrease reduces the likelihood of forming hetero-tetramers, in which the isoforms will be in higher proportion than FLp53. Thus, it seems unlikely that hetero-tetramerization of $\Delta 133p53$ and $\Delta 160p53$ isoforms with FLp53 would be the main reason for their dominant-negative effect.

Next we investigated whether aggregation of the $\Delta 133p53$ and $\Delta 160p53$ isoforms contributes to their functional loss and dominant-negative effect on FLp53. Our immunofluorescence and immunoprecipitation analyses of H1299 cells expressing these isoforms show a higher aggregation propensity of the two isoforms than FLp53 (**Figure 5C** and Supplementary Figure 2). The aggregation likely results from the exposure of hydrophobic sequences in the isoforms (**Figure 1C**). In addition, the absence of N-terminal TADs in $\Delta 133p53$ and $\Delta 160p53$ isoforms may prevent their MDM2-mediated ubiquitin-proteasome degradation^{66, 76}. This may, in turn, lead to higher accumulation of $\Delta 133p53$ and $\Delta 160p53$ isoforms in the cell, which, combined with an unstable core domain, promotes their aggregation. Moreover, the observed cytoplasmic isoform aggregates may reflect autophagy-related degradation, as suggested by the co-localization of $\Delta 133p53$ with autophagy substrate p62/SQSTM1 and autophagosome component LC3B⁷⁷. Overexpression of these aggregation-prone proteins could induce endoplasmic reticulum stress and activate autophagy⁷⁸. Interestingly, we also observed nuclear aggregation of these isoforms (**Figure 6B** and **E** and Supplementary Figure 6B), suggesting that distinct mechanisms, such as intrinsic properties of the isoforms, may govern their localization and behavior within the nucleus. This dual localization underscores the complexity of $\Delta 133p53$ and $\Delta 160p53$ behavior in cellular systems.

More interestingly, our immunofluorescence and immunoprecipitation assays demonstrate large co-aggregation of FLp53 with $\Delta 133p53$ and $\Delta 160p53$ isoforms in the cytoplasm as well as in the nucleus when expressed together (**Figure 6E** and Supplementary Figure 4, panels 3 and 4). The predominant cytoplasmic aggregation would sequester FLp53 in the cytoplasm. Moreover, it may

destabilize FLp53's conformation, potentially masking the nuclear localization sequence (NLS) and preventing its nuclear import⁷⁹. Additionally, it may disrupt the interaction of FLp53 with MDM2, thereby inhibiting its MDM2-induced degradation and enhancing its accumulation⁸⁰, which would drive it more to the aggregation pathway. All these processes involving FLp53 aggregation would result in reduction of the functional FLp53 in the cell and more importantly, in the nucleus. Cytoplasmic aggregation would also limit the isoforms from nuclear import, reducing the possibility of hetero-tetramerization. Therefore, we propose that the co-aggregation of $\Delta 133$ p53 and $\Delta 160$ p53 with FLp53 is the primary factor for their dominant-negative effect. This mechanism is consistent with the model where structural mutations, such as R175H and R282W in FLp53, induce co-aggregation and exert a dominant-negative effect⁴⁰.

In our study, we primarily utilized an overexpression strategy involving FLAG/V5-tagged proteins to investigate the effects of p53 isoforms $\Delta 133$ p53 and $\Delta 160$ p53 on the function of FLp53. To address concerns regarding potential overexpression artifacts, we performed the co-immunoprecipitation (Supplementary Figure 6) and caspase-3 and -7 activity (Figure 7) experiments with untagged $\Delta 133$ p53 and $\Delta 160$ p53. In both experimental systems, the untagged proteins behaved very similarly to the FLAG/V5 antibody epitope-containing proteins (Figures 6 and 7 and Supplementary Figure 6). Hence, the C-terminal tagging of FLp53 or its isoforms does not alter the biochemical and physiological functions of these proteins.

We propose a model to explain how the p53 isoforms $\Delta 133$ p53 and $\Delta 160$ p53 inhibit FLp53 activity and exert a dominant-negative effect on it (Figure 8A). Two mechanisms are involved: a subsidiary mechanism in which $\Delta 133$ p53 and $\Delta 160$ p53 isoforms form hetero-tetramers with FLp53 impairing its DNA-binding and transcriptional activities, and a predominant mechanism in which these isoforms promote co-aggregation with FLp53 and abolish its transcriptional activities by disrupting its structure, regulation and degradation pathways. Since elevated levels of $\Delta 133$ p53 and $\Delta 160$ p53 relative to FLp53 attenuate its transcriptional activity (Figure 8B, left) and overexpression of FLp53 relative to the isoforms allows to regain it (Figure 8B, right), the tumor-suppressive efficacy of p53 must depend on the balanced expression of FLp53 and its isoforms, highlighting the importance of their relative abundance in cancer cells. Based on our results and recent related publications^{38, 39}, we propose that co-aggregation, rather than hetero-tetramerization, is the primary mechanism for the dominant-negative effect exerted by the p53 isoforms with large N-terminal truncations.

Methods

Cell culture, plasmids, and plasmid transfections

Human non-small cell lung carcinoma cells H1299 (p53-null) were obtained from ATCC and cultured in Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% fetal bovine serum (FBS, Thermo Fisher Scientific) and penicillin-streptomycin solution (PEST; Thermo Fisher Scientific) at 37 °C in a 5% CO₂ incubator. The plasmid pcDNA3.1-FLp53-FLAG, encoding human full-length p53 (amino acids 1-393), was synthesized by GenScript (Rijswijk, Netherlands). A flexible linker (GGGGG)₁ was introduced between the FLp53 coding sequence and the C-terminal FLAG epitope tag. Subsequently, this fusion construct was inserted into the pcDNA3.1 vector at the NheI and EcoRI restriction enzyme sites. The plasmids encoding N-terminally truncated p53 proteins ($\Delta 133$ p53 and $\Delta 160$ p53) were created by deleting the N-terminal 132 and 159 amino acids from the FLp53 sequence, respectively. To create plasmids expressing C-terminal V5-tagged proteins (FLp53, $\Delta 133$ p53, and $\Delta 160$ p53), the respective DNA sequences were amplified by PCR from the pcDNA3.1-FLp53-FLAG template and inserted into the EcoRI and XhoI sites of the pcDNA3.1-V5. The plasmids encoding untagged p53 proteins were constructed by inserting the respective DNA sequences into the NheI and EcoRI restriction sites of the pcDNA3.1 empty vector. Luciferase reporter plasmids, WWP/p21-Luc (Plasmid #16451)⁶, pGL3-MDM2-Luc (Plasmid

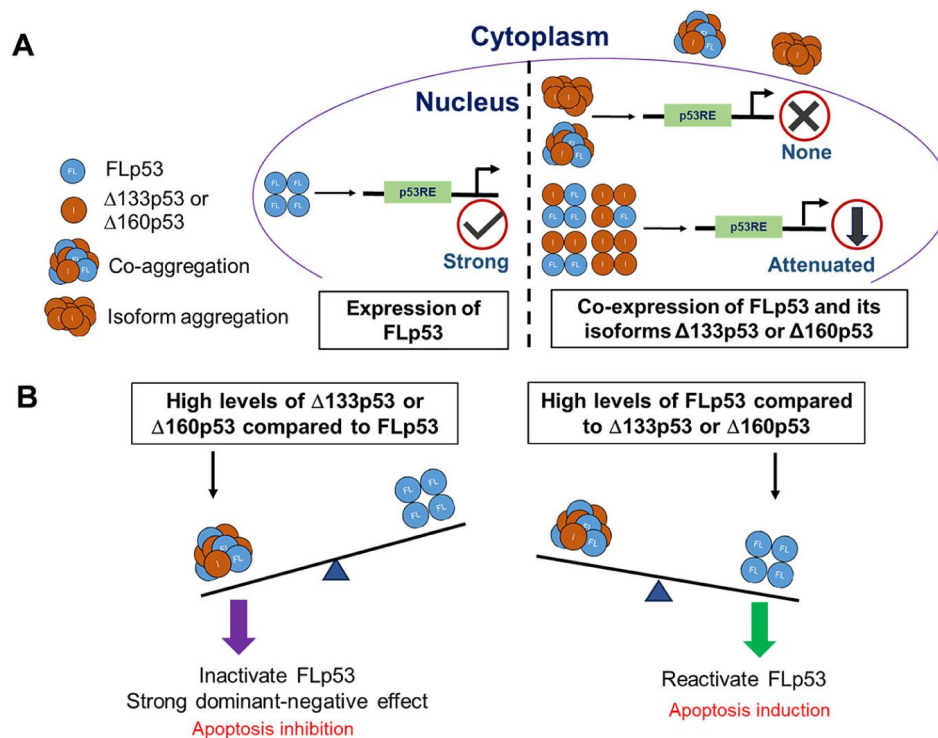


Figure 8.

(A) Schematic representation of the underlying mechanisms by which $\Delta 133p53$ and $\Delta 160p53$ exert dominant-negative effect on FLp53.

(A) When expressed alone, FLp53 demonstrates strong transcriptional activity, but when expressed together with the $\Delta 133p53$ or $\Delta 160p53$ isoforms, FLp53 activity gets impaired due to the dominant-negative effect exerted by the isoforms. The $\Delta 133p53$ or $\Delta 160p53$ isoforms either attenuate FLp53's transcriptional activity by hetero-tetramerization in a ratio higher than 1:1, or abolish it completely by aggregation. Since $\Delta 133p53$ and $\Delta 160p53$ exhibit much higher propensity for aggregation than FLp53, they sequester FLp53 in the cytoplasmic aggregates, leading to reduction in the amount of functional FLp53 in the nucleus. Aggregation also limits the availability of these isoforms in the nucleus. Thus, co-aggregation is the primary reason for dominant negative effect of the $\Delta 133p53$ or $\Delta 160p53$ isoforms on FLp53. It is important to note that the numbers of tetramers and aggregates depicted do not represent the actual quantities present in the cytoplasm and nucleus. **(B)** The balance between the expression level of the FLp53 and its isoforms $\Delta 133p53$ and $\Delta 160p53$ regulates FLp53 activity. Increased levels of $\Delta 133p53$ and $\Delta 160p53$ relative to FLp53, lead to stronger dominant-negative effect due to co-aggregation and hetero-tetramerization, resulting in the inhibition of apoptosis. Increased levels of FLp53 compared to the isoforms $\Delta 133p53$ and $\Delta 160p53$ in cancer cells restore the function of FLp53, leading to the induction of apoptosis.

#32365)⁸¹[\[link\]](#), and PUMA Frag1-Luc (Plasmid #16591)⁸²[\[link\]](#) were obtained from Addgene. For the construction of the BAX-Luc reporter plasmid, the p53 binding site in the BAX promoter was PCR amplified from HepG2 genomic DNA kindly provided by Dr. Gang Pan (Uppsala University) and cloned into the pGL3-Basic plasmid (Promega) using XhoI and HindIII sites. The PCR primers used for cloning are listed in Supplementary Table 1. All plasmid transfections were performed with jetPRIME[®] (Polyplus) transfection reagent according to the manufacturer's protocol.

Immunoprecipitation and Western blotting

H1299 cells were seeded in a 6-well plate and transfected with 2 µg of various plasmids. The total amount of plasmids (2 µg/well) was kept constant in all samples by adding empty plasmid pcDNA3.1. To investigate p53 oligomerization, cells were co-transfected with equal amounts (1 µg each) of FLAG- and V5-tagged plasmids. After 24 h post-transfection (hpt), cells were harvested and lysed in RIPA buffer (50 mM Tris-HCl pH 7.4, 150 mM NaCl, 1% Triton X-100, 1% sodium deoxycholate, 0.1% SDS, and 1 mM EDTA) supplemented with the Halt Protease Inhibitor Cocktail (Thermo Fischer Scientific) as previously described⁸³[\[link\]](#). The whole cell lysates were incubated for 30 min on ice and centrifuged at 13,000 × g for 15 min at 4°C. A fraction of the cleared cell lysates was reserved as the input sample; the remaining lysates were used for immunoprecipitation with anti-FLAG M2 magnetic beads (Sigma, Cat No. M8823) overnight at 4°C. The beads were washed with 4 × 1 ml of RIPA buffer. Proteins were then eluted from the beads using 2 × Laemmli Sample buffer, separated on a 10% SDS-PAGE, transferred onto nitrocellulose membranes (Bio-Rad), and blocked with 5% nonfat dry milk in TBS containing 0.1% Tween-20 (TBST) for 1 h at room temperature. The blocked membranes were incubated overnight at 4°C with the anti-FLAG (Sigma, Cat No. F2555) and anti-V5 (Sigma, Cat No. V8012) primary antibodies in a blocking buffer overnight at 4°C. Proteins were detected with the IRDye 680RD Goat anti-Rabbit IgG and IRDye 800CW Donkey anti-Mouse IgG secondary antibodies (Li-COR Biosciences). The Western blots were visualized using an Odyssey CLX imaging system (Li-COR Biosciences).

Chromatin immunoprecipitation (ChIP) assays

H1299 cells were seeded in 10 cm cell culture dishes and transfected with 5 µg of either pcDNA3.1-FLp53-FLAG, pcDNA3.1-Δ133p53-FLAG, pcDNA-Δ160p53-FLAG or an empty vector pcDNA3.1 for 24 h. To investigate the impacts of isoforms Δ133p53 and Δ160p53 on the DNA-binding activity of FLp53, H1299 cells were co-transfected with 2.5 µg of pcDNA3.1-FLp53-FLAG along with either 2.5 µg of pcDNA3.1-Δ133p53-V5, pcDNA3.1-Δ160p53-V5 or an equal amount of empty vector pcDNA3.1. The ChIP assays were performed as described previously⁸⁴[\[link\]](#), with minor modification. Briefly, cells were fixed with 1% formaldehyde in PBS for 10 min at room temperature, followed by lysis in 250 µl of lysis buffer (50 mM Tris-HCl pH 8.0, 10 mM EDTA, 1% SDS) supplemented with Halt Protease Inhibitor Cocktail. Lysates were then sonicated for 15 cycles (30 s ON/30 s OFF) using the Bioruptor Pico (Diagenode). The resulting soluble cell lysates were diluted with dilution buffer (16.7 mM Tris-HCl pH 8.0, 167 mM NaCl, 1.2 mM EDTA, 0.01% SDS, 1.1% Triton X-100) and immunoprecipitated overnight at 4°C with the anti-FLAG M2 magnetic beads. After a series of washes, protein-DNA complexes were eluted, and the crosslinks were reversed through incubation with RNase A (Thermo Fischer Scientific) at 37°C for 30 min, followed by proteinase K (Invitrogen) digestion at 65°C overnight. The enrichment of specific DNA fragments was evaluated with qPCR analysis using HOT FIREPol EvaGreen qPCR Supermix (Solis Biodyne). All qPCR reactions were done in triplicate, employing the QuantStudio[™] 6 Flex Real-Time PCR System (Thermo Fisher Scientific). The relative fold enrichments were calculated after normalizing to the input values and considering the pcDNA3.1 transfected samples as 1. Primer sequences used for qPCR analysis are listed in Supplementary Table 1.

Luciferase assay

H1299 cells were seeded in 24-well plates and transfected in triplicate with 25 ng/well of luciferase reporter plasmids and various p53-expressing plasmids. Co-transfection ratios for FLp53 and its isoforms $\Delta 133p53$ or $\Delta 160p53$, were set at 1:1 (12.5 ng/well of FLp53 to 12.5 ng/well of isoform), 1:5 (12.5 ng/well of FLp53 to 62.5 ng/well of isoform), and 1:10 (12.5 ng/well of FLp53 to 125 ng/well of isoform). The total plasmid amount (300 ng/well) was kept consistent across the various groups by supplementing with the empty pcDNA3.1 plasmid. Cells were harvested 24 hpt and lysed in a Passive Lysis Buffer (Promega). The cell lysate was centrifuged at $13,000 \times g$ for 5 min at 4°C , and 20 μl of the lysate was analyzed in triplicates using a firefly luciferase substrate on Infinite 200 PRO plate reader (TECAN)⁸⁵. The luciferase activity of the FLp53 and its isoforms transfected samples was normalized against that of samples transfected with the control vector pcDNA3.1 to obtain relative reporter activity. Results represent the mean values and standard deviations of three independent experimental replicates. The data was analyzed using Origin 2018 software (Origin Lab Corporation, USA).

Subcellular fractionation

The subcellular fractionation assay was carried out using NE-PER Nuclear and Cytoplasmic Extraction Reagents (Thermo Fischer Scientific), with minor modifications. In brief, H1299 cells were seeded in 10 cm cell culture dishes and transfected with 0.8 μg of the following plasmids: pcDNA3.1-FLp53-FLAG, pcDNA3.1- $\Delta 133p53$ -FLAG, pcDNA- $\Delta 160p53$ -FLAG, and the control vector pcDNA3.1. To investigate the effects of overexpressing p53 isoforms on the subcellular localization of FLp53, co-transfection was performed using FLAG-tagged FLp53 and V5-tagged $\Delta 133p53$ or $\Delta 160p53$ at a 1:5 ratio (0.8 μg to 4.2 μg). Cells were transfected with 5 μg of the empty vector pcDNA3.1 as a control. About 24 hpt, cells were collected by trypsinization, resuspended in ice-cold CER I buffer containing Halt Protease Inhibitor Cocktail and incubated for 10 min on ice. Subsequently, ice-cold CER II buffer was added according to the manufacturer's instructions. The cytoplasmic extracts were isolated by centrifugation at $14,000 \times g$ for 5 min at 4°C . The nuclear pellet was then resolved in ice-cold RIPA buffer supplemented with Halt Protease Inhibitor Cocktail, subjected to vortex for 15 s every 10 min for a total of 40 min, and centrifuged at $14,000 \times g$ for 10 min to obtain the soluble nuclear fraction. The insoluble nuclear pellet (i.e., insoluble nuclear fraction) was resuspended in RIPA buffer, an appropriate volume of $2 \times$ Laemmli Sample buffer was added, and the samples were heated at 95°C until the pellet fully dissolved. For Western blot experiments, equal volumes of each subcellular fraction were separated by SDS-PAGE, and the nitrocellulose membranes were probed with the respective antibodies. Histone H3 (nuclear marker) and α -Tubulin (cytoplasmic marker) antibodies were used to verify the purity of the nuclear and cytoplasmic fractions. All primary antibodies: anti-FLAG (Sigma, Cat No. F2555), anti-V5 (Sigma, Cat No. V8012 and Cat No. AB3792), anti-Histone H3 (Abcam, Cat No. ab1791), and anti- α -Tubulin (Sigma, Cat No. T6199) were incubated on membranes overnight at 4°C . The Western blots were visualized using an Odyssey CLX imaging system, and the proteins were quantified using Image Studio Version 5.5 (Li-COR).

Immunofluorescence analysis

H1299 cells were seeded on glass coverslips in 24-well plates and transfected with 250 ng/well of either pcDNA3.1-FLp53-FLAG, pcDNA3.1- $\Delta 133p53$ -FLAG, pcDNA- $\Delta 160p53$ -FLAG or the control vector pcDNA3.1. To investigate whether an excess of p53 isoforms could influence the subcellular localization of FLp53, H1299 cells were co-transfected with 250 ng/well of pcDNA3.1-FLp53-FLAG along with 1000 ng/well of pcDNA3.1- $\Delta 133p53$ -V5, pcDNA3.1- $\Delta 160p53$ -V5, or pcDNA3.1. About 24 hpt, cells were fixed with 4% paraformaldehyde in PBS, followed by permeabilization with 0.2% Triton X-100 in PBS for 15 min at room temperature as described previously⁸⁶. Fixed cells on coverslips were incubated overnight at 4°C with primary antibodies anti-FLAG (Sigma, Cat No 1804, 1:1000 dilution), and anti-V5 (Sigma, Cat No AB3792, 1:1000 dilution). Subsequently,

coverslips were incubated with the Alexa Fluor 488 goat anti-mouse IgG (Invitrogen, Cat No. A11001, 1:2000 dilution) and/or Alexa Fluor 594 chicken anti-rabbit IgG (H+L) (Invitrogen, Cat No. A21442, 1:2000 dilution) secondary antibodies for 30 min at room temperature. The coverslips were mounted with ProLong Diamond Antifade Mountant (Invitrogen, Cat No. P36961), and nuclei were counterstained using DAPI. Subcellular localization was visualized using a Nikon Eclipse 90i microscope (Nikon, Japan). Images were processed using ImageJ software⁸⁷ and subsequently arranged by Inkscape.

Caspase-3/7 activity assay

Caspase-3 and -7 activities were measured with a Caspase-Glo® 3/7 assay kit (Promega, Madison, USA) according to the manufacturer's protocol. Briefly, H1299 cells were seeded (5000 cells/well) into a white solid bottom 96-well plate and transfected in triplicate with 180 ng/well of plasmids expressing FLAG-tagged or untagged FLp53, Δ133p53, Δ160p53, or the empty vector control pcDNA3.1. To evaluate the effects of overexpression of isoforms on the apoptotic activity of FLp53, H1299 cells were co-transfected with plasmids encoding FLp53 (30 ng/well) with the control plasmid pcDNA3.1 (150 ng/well), or plasmids expressing isoforms Δ133p53 and Δ160p53 (150 ng/well), at a 1:5 ratio. After 28 hpt, caspase-3/7 substrate was added to each well and incubated for 90 min at room temperature. Luminescence was recorded using an Infinite 200 PRO plate reader (TECAN). The background luminescence associated with the cell culture and assay reagent (blank reaction) was subtracted from the experimental values.

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Additional information

Author contribution

S.S. conceived the idea of the project; L.Z. and T.P. performed experiments; L.Z., T.P., and S.S. analyzed the data; L.Z. wrote the manuscript and prepared Figures. All authors reviewed, edited, and approved the manuscript.

Additional files

Supplementary [↗](#)

Review response [↗](#)

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Reviewer #1 (Public review):

Summary:

The authors have provided a mechanism by which how presence of truncated P53 can inactivate function of full length P53 protein. The authors proposed this happens by sequestration of full length P53 by truncated P53. In the study, the performed experiments are well described.

Significance:

The work is significant, since it points out more mechanistic insight how wild type full length P53 could be inactivated in the presence of truncated isoforms, this might offer new opportunity to recover P53 function as treatment strategies against cancer.

Comments on latest version:

The authors have made significant effort to address my concerns using the system available to them. I find the justifications provided in the rebuttal letter and the revised figures satisfactory. My initial concerns regarding the overexpression system have been largely addressed. However, the experimental system used by the authors lacks the means to measure the effect on endogenous p53, which remains a limitation.

<https://doi.org/10.7554/eLife.106469.1.sa2>

Reviewer #2 (Public review):

Summary:

The revised manuscript by Zhao and colleagues presents a novel and compelling investigation into the p53 isoforms, $\Delta 133p53$ and $\Delta 160p53$, which are implicated in aggressive cancer phenotypes. The primary goal of this study was to elucidate how these isoforms exert a dominant-negative impact on the activity of full-length p53 (FLp53). The authors demonstrate that the $\Delta 133p53$ and $\Delta 160p53$ isoforms display impaired binding to p53-regulated promoters. Their findings suggest that the dominant-negative effects observed are primarily due to the co-aggregation of FLp53 with $\Delta 133p53$ and $\Delta 160p53$.

Overall, the study is innovative, thoroughly executed, and supported by robust data analysis. The authors have effectively addressed the reviewers' criticisms and incorporated their suggestions in this revised manuscript.

Significance:

The manuscript by Zhao and colleagues presents a novel and compelling study on the p53 isoforms, $\Delta 133p53$ and $\Delta 160p53$, which are associated with aggressive cancer types. The main

objective of the study was to understand how these isoforms exert a dominant negative effect on full-length p53 (FLp53). The authors discovered that the $\Delta 133$ p53 and $\Delta 160$ p53 proteins exhibit impaired binding to p53-regulated promoters. The data suggest that the predominant mechanism driving the dominant-negative effect is the co-aggregation of FLp53 with $\Delta 133$ p53 and $\Delta 160$ p53.

<https://doi.org/10.7554/eLife.106469.1.sa1>

Author response:

Reviewer #1 (Evidence, reproducibility and clarity):

Authors has provided a mechanism by which how presence of truncated P53 can inactivate function of full length P53 protein. Authors proposed this happens by sequestration of full length P53 by truncated P53.

In the study, performed experiments are well described.

My area of expertise is molecular biology/gene expression, and I have tried to provide suggestions on my area of expertise. The study has been done mainly with overexpression system and I have included few comments which I can think can be helpful to understand effect of truncated P53 on endogenous wild type full length protein. Performing experiments on these lines will add value to the observation according to this reviewer.

Major comments:

(1) What happens to endogenous wild type full length P53 in the context of mutant/truncated isoforms, that is not clear. Using a P53 antibody which can detect endogenous wild type P53, can authors check if endogenous full length P53 protein is also aggregated as well? It is hard to differentiate if aggregation of full length P53 happens only in overexpression scenario, where lot more both of such proteins are expressed. In normal physiological condition P53 expression is usually low, tightly controlled and its expression get induced in altered cellular condition such as during DNA damage. So, it is important to understand the physiological relevance of such aggregation, which could be possible if authors could investigate effect on endogenous full length P53 following overexpression of mutant isoforms.

Thank you very much for your insightful comments.

(1) To address “what happens to endogenous wild-type full-length P53 in the context of mutant/truncated isoforms,” we employed a human A549 cell line expressing endogenous wild-type p53 under DNA damage conditions such as an etoposide treatment(1). We choose the A549 cell line since similar to H1299, it is a lung cancer cell line (www.atcc.org). For comparison, we also transfected the cells with 2 μ g of V5-tagged plasmids encoding FLp53 and its isoforms $\Delta 133$ p53 and $\Delta 160$ p53. As shown in Author response image 1A, lanes 1 and 2, endogenous p53 expression, remained undetectable in A549 cells despite etoposide treatment, which limits our ability to assess the effects of the isoforms on the endogenous wild-type FLp53. We could, however, detect the V5-tagged FLp53 expressed from the plasmid using anti-V5 (rabbit) as well as with antiDO-1 (mouse) antibody (Author response image 1). The latter detects both endogenous wildtype p53 and the V5-tagged FLp53 since the antibody epitope is within the Nterminus (aa 20-25). This result supports the reviewer’s comment regarding the low level of expression of endogenous p53 that is insufficient for detection in our experiments.

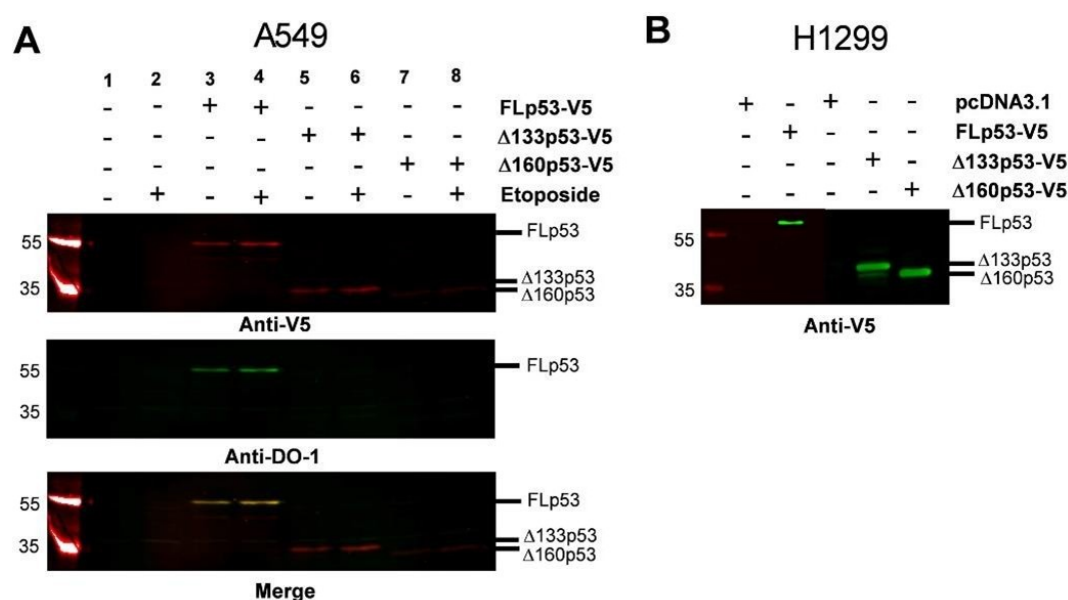
In summary, in line with the reviewer's comment that 'under normal physiological conditions p53 expression is usually low,' we could not detect p53 with an anti-DO-1 antibody. Thus, we proceeded with V5/FLAG-tagged p53 for detection of the effects of the isoforms on p53 stability and function. We also found that protein expression in H1299 cells was more easily detectable than in A549 cells (Compare Author response image 1A and B). Thus, we decided to continue with the H1299 cells (p53-null), which would serve as a more suitable model system for this study.

(2) We agree with the reviewer that 'It is hard to differentiate if aggregation of full-length p53 happens only in overexpression scenario'. However, it is not impossible to imagine that such aggregation of FLp53 happens under conditions when p53 and its isoforms are over-expressed in the cell. Although the exact physiological context is not known and beyond the scope of the current work, our results indicate that at higher expression, p53 isoforms drive aggregation of FLp53. Given the challenges of detecting endogenous FLp53, we had to rely on the results obtained with plasmid mediated expression of p53 and its isoforms in p53-null cells.

Author response image 1.

Comparative analysis of protein expression in A549 and H1299 cells. (A) A549 cells (p53 wild-type) were treated with etoposide to induce endogenous wild-type p53 expression. To assess the effects of FLp53 and its isoforms $\Delta 133p53$ and $\Delta 160p53$ on endogenous wild-type p53 aggregation, A549 cells were transfected with 2 μ g of V5-tagged p53 expression plasmids, with or without etoposide (20 μ M for 8h) treatment. Western blot analysis was done with the anti-V5 (rabbit) to detect V5-tagged proteins and anti-DO-1 (mouse), the latter detects both endogenous wild-type p53 and V5-tagged FLp53. The merged image corresponds to the overlay between the V5 and DO1 antibody signals. (B) H1299 cells (p53-null) were transfected with 2 μ g V5tagged p53 expression plasmids or the empty vector control pcDNA3.1.

Western blot analysis was done with the anti-V5 (mouse) antibody.



(2) Can presence of mutant P53 isoforms can cause functional impairment of wild type full length endogenous P53? That could be tested as well using similar ChIP assay authors has performed, but instead of antibody against the Tagged protein if the

authors could check endogenous P53 enrichment in the gene promoter such as P21 following overexpression of mutant isoforms. May be introducing a condition such as DNA damage in such experiment might help where endogenous P53 is induced and more prone to bind to P53 target such as P21.

Thank you very much for your valuable comments and suggestions. To investigate the potential functional impairment of endogenous wild-type p53 by p53 isoforms, we initially utilized A549 cells (p53 wild-type), aiming to monitor endogenous wild-type p53 expression following DNA damage. However, as mentioned and demonstrated in Author response image 1, endogenous p53 expression was too low to be detected under these conditions, making the ChIP assay for analyzing endogenous p53 activity unfeasible. Thus, we decided to utilize plasmid-based expression of FLp53 and focus on the potential functional impairment induced by the isoforms.

(3) On similar lines, authors described:

"To test this hypothesis, we escalated the ratio of FLp53 to isoforms to 1:10. As expected, the activity of all four promoters decreased significantly at this ratio (Figure 4A-D). Notably, Δ160p53 showed a more potent inhibitory effect than Δ133p53 at the 1:5 ratio on all promoters except for the p21 promoter, where their impacts were similar (Figure 4E-H). However, at the 1:10 ratio, Δ133p53 and Δ160p53 had similar effects on all transactivation except for the MDM2 promoter (Figure 4E-H)."

Again, in such assay authors used ratio 1:5 to 1:10 full length vs mutant. How authors justify this result in context (which is more relevant context) where one allele is Wild type (functional P53) and another allele is mutated (truncated, can induce aggregation). In this case one would expect 1:1 ratio of full-length vs mutant protein, unless other regulation is going which induces expression of mutant isoforms more than wild type full length protein. Probably discussing on these lines might provide more physiological relevance to the observed data.

Thank you for raising this point regarding the physiological relevance of the ratios used in our study.

(1) In the revised manuscript (lines 193-195), we added in this direction that "The elevated Δ133p53 protein modulates p53 target genes such as miR-34a and p21, facilitating cancer development(2, 3). To mimic conditions where isoforms are upregulated relative to FLp53, we increased the ratios to 1:5 and 1:10." This approach aims to simulate scenarios where isoforms accumulate at higher levels than FLp53, which may be relevant in specific contexts, as also elaborated above.

(2) Regarding the issue of protein expression, where one allele is wild-type and the other is isoform, this assumption is not valid in most contexts. First, human cells have two copies of TPp53 gene (one from each parent). Second, the TP53 gene has two distinct promoters: the proximal promoter (P1) primarily regulates FLp53 and Δ40p53, whereas the second promoter (P2) regulates Δ133p53 and Δ160p53(4, 5). Additionally, Δ133TP53 is a p53 target gene(6, 7) and the expression of Δ133p53 and FLp53 is dynamic in response to various stimuli. Third, the expression of p53 isoforms is regulated at multiple levels, including transcriptional, post-transcriptional, translational, and post-translational processing(8). Moreover, different degradation mechanisms modify the protein level of p53 isoforms and FLp53(8). These differential regulation mechanisms are regulated by various stimuli, and therefore, the 1:1 ratio of FLp53 to Δ133p53 or Δ160p53 may be valid only under certain physiological conditions. In line with this, varied expression levels of FLp53 and its isoforms, including Δ133p53 and Δ160p53, have been reported in several studies(3, 4, 9, 10).

(3) In our study, using the pcDNA 3.1 vector under the human cytomegalovirus (CMV) promoter, we observed moderately higher expression levels of $\Delta 133p53$ and $\Delta 160p53$ relative to FLp53 (Author response image 1B). This overexpression scenario provides a model for studying conditions where isoform accumulation might surpass physiological levels, impacting FLp53 function. By employing elevated ratios of these isoforms to FLp53, we aim to investigate the potential effects of isoform accumulation on FLp53.

(4) Finally does this altered function of full length P53 (preferably endogenous one) in presence of truncated P53 has any phenotypic consequence on the cells (if authors choose a cell type which is having wild type functional P53). Doing assay such as apoptosis/cell cycle could help us to get this visualization.

Thank you for your insightful comments. In the experiment with A549 cells (p53 wild-type), endogenous p53 levels were too low to be detected, even after DNA damage induction. The evaluation of the function of endogenous p53 in the presence of isoforms is hindered, as mentioned above. In the revised manuscript, we utilized H1299 cells with overexpressed proteins for apoptosis studies using the Caspase-Glo® 3/7 assay (Figure 7). This has been shown in the Results section (lines 254-269). “The $\Delta 133p53$ and $\Delta 160p53$ proteins block pro-apoptotic function of FLp53.

One of the physiological read-outs of FLp53 is its ability to induce apoptotic cell death(11). To investigate the effects of p53 isoforms $\Delta 133p53$ and $\Delta 160p53$ on FLp53-induced apoptosis, we measured caspase-3 and -7 activities in H1299 cells expressing different p53 isoforms (Figure 7). Caspase activation is a key biochemical event in apoptosis, with the activation of effector caspases (caspase-3 and -7) ultimately leading to apoptosis(12). The caspase-3 and -7 activities induced by FLp53 expression was approximately 2.5 times higher than that of the control vector (Figure 7). Co-expression of FLp53 and the isoforms $\Delta 133p53$ or $\Delta 160p53$ at a ratio of 1: 5 significantly diminished the apoptotic activity of FLp53 (Figure 7). This result aligns well with our reporter gene assay, which demonstrated that elevated expression of $\Delta 133p53$ and $\Delta 160p53$ impaired the expression of apoptosis-inducing genes BAX and PUMA (Figure 4G and H). Moreover, a reduction in the apoptotic activity of FLp53 was observed irrespective of whether $\Delta 133p53$ or $\Delta 160p53$ protein was expressed with or without a FLAG tag (Figure 7). This result, therefore, also suggests that the FLAG tag does not affect the apoptotic activity or other physiological functions of FLp53 and its isoforms. Overall, the overexpression of p53 isoforms $\Delta 133p53$ and $\Delta 160p53$ significantly attenuates FLp53-induced apoptosis, independent of the protein tagging with the FLAG antibody epitope.”

Referees cross-commenting

I think the comments from the other reviewers are very much reasonable and logical.

Especially all 3 reviewers have indicated, a better way to visualize the aggregation of full-length wild type P53 by truncated P53 (such as looking at endogenous P53# by reviewer 1, having fluorescent tag #by reviewer 2 and reviewer 3 raised concern on the FLAG tag) would add more value to the observation.

Thank you for these comments. The endogenous p53 protein was undetectable in A549 cells induced by etoposide (Figure R1A). Therefore, we conducted experiments using FLAG/V5-tagged FLp53. To avoid any potential side effects of the FLAG tag on p53 aggregation, we introduced untagged p53 isoforms in the H1299 cells and performed subcellular fractionation. Our revised results, consistent with previous FLAG-tagged p53 isoforms findings, demonstrate that co-expression of untagged isoforms with FLAG-tagged FLp53 significantly induced the aggregation of FLAG-FLp53, while no aggregation was observed

when FLAG-tagged FLp53 was expressed alone (Supplementary Figure 6). These results clearly indicate that the FLAG tag itself does not contribute to protein aggregation.

Additionally, we utilized the A11 antibody to detect protein aggregation, providing additional validation (Figure 8 from Jean-Christophe Bourdon et al. *Genes Dev.* 2005;19:2122-2137). Given that the fluorescent proteins (~30 kDa) are substantially bigger than the tags used here (~1 kDa) and may influence oligomerization (especially GFP), stability, localization, and function of p53 and its isoforms, we avoided conducting these vital experiments with such artificial large fusions.

Reviewer #1 (Significance):

The work is significant, since it points out more mechanistic insight how wild type full length P53 could be inactivated in the presence of truncated isoforms, this might offer new opportunity to recover P53 function as treatment strategies against cancer.

Thank you for your insightful comments. We appreciate your recognition of the significance of our work in providing mechanistic insights into how wild-type FLp53 can be inactivated by truncated isoforms. We agree that these findings have potential for exploring new strategies to restore p53 function as a therapeutic approach against cancer.

Reviewer #2 (Evidence, reproducibility and clarity):

The manuscript by Zhao and colleagues presents a novel and compelling study on the p53 isoforms, $\Delta 133p53$ and $\Delta 160p53$, which are associated with aggressive cancer types. The main objective of the study was to understand how these isoforms exert a dominant negative effect on full-length p53 (FLp53). The authors discovered that the $\Delta 133p53$ and $\Delta 160p53$ proteins exhibit impaired binding to p53-regulated promoters. The data suggest that the predominant mechanism driving the dominant-negative effect is the coaggregation of FLp53 with $\Delta 133p53$ and $\Delta 160p53$.

This study is innovative, well-executed, and supported by thorough data analysis. However, the authors should address the following points:

(1) Introduction on Aggregation and Co-aggregation: Given that the focus of the study is on the aggregation and co-aggregation of the isoforms, the introduction should include a dedicated paragraph discussing this issue. There are several original research articles and reviews that could be cited to provide context.

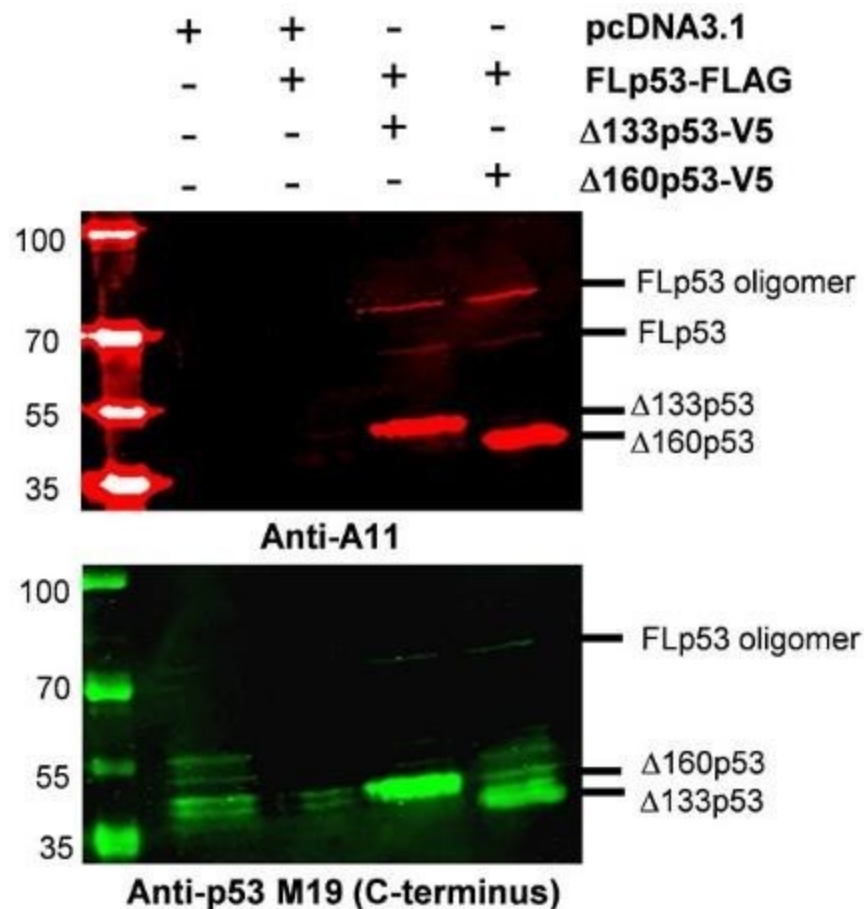
Thank you very much for the valuable comments. We have added the following paragraph in the revised manuscript (lines 74-82): “Protein aggregation has become a central focus of modern biology research and has documented implications in various diseases, including cancer(13, 14, 15). Protein aggregates can be of different types ranging from amorphous aggregates to highly structured amyloid or fibrillar aggregates, each with different physiological implications. In the case of p53, whether protein aggregation, and in particular, co-aggregation with large N-terminal deletion isoforms, plays a mechanistic role in its inactivation is yet underexplored. Interestingly, the $\Delta 133p53\beta$ isoform has been shown to aggregate in several human cancer cell lines(16). Additionally, the $\Delta 40p53\alpha$ isoform exhibits a high aggregation tendency in endometrial cancer cells(17). Although no direct evidence exists for $\Delta 160p53$ yet, these findings imply that p53 isoform aggregation may play a major role in their mechanisms of actions.”

(2) Antibody Use for Aggregation: To strengthen the evidence for aggregation, the authors should consider using antibodies that specifically bind to aggregates.

Thank you for your insightful suggestion. We addressed protein aggregation using the A11 antibody which specifically recognizes amyloid-like protein aggregates. We analyzed insoluble nuclear pellet samples prepared under identical conditions as described in Figure 6B. To confirm the presence of p53 proteins, we employed the anti-p53 M19 antibody (Santa Cruz, Cat No. sc-1312) to detect bands corresponding to FLp53 and its isoforms $\Delta 133p53$ and $\Delta 160p53$. The monomer FLp53 was not detected (Figure 8, lower panel, Jean-Christophe Bourdon et al. *Genes Dev.* 2005;19:2122-2137), which may be attributed to the lower binding affinity of the anti-p53 M19 antibody to it. These samples were also immunoprecipitated using the A11 antibody (Thermo Fischer Scientific, Cat No. AHB0052) to detect aggregated proteins. Interestingly, FLp53 and its isoforms, $\Delta 133p53$ and $\Delta 160p53$, were clearly visible with Anti-A11 antibody when co-expressed at a 1:5 ratio suggesting that they underwent co-aggregation. However, no FLp53 aggregates were observed when it was expressed alone (Author response image 2). These results support the conclusion in our manuscript that $\Delta 133p53$ and $\Delta 160p53$ drive FLp53 aggregation.

Author response image 2.

Induction of FLp53 Aggregation by p53 Isoforms $\Delta 133p53$ and $\Delta 160p53$. H1299 cells transfected with the FLAG-tagged FLp53 and V5-tagged $\Delta 133p53$ or $\Delta 160p53$ at a 1:5 ratio. The cells were subjected to subcellular fractionation, and the resulting insoluble nuclear pellet was resuspended in RIPA buffer. The samples were heated at 95°C until the pellet was completely dissolved, and then analyzed by Western blotting. Immunoprecipitation was performed using the A11 antibody, which specifically recognizes amyloid protein aggregates, and the anti-p53 M19 antibody, which detects FLp53 as well as its isoforms $\Delta 133p53$ and $\Delta 160p53$.



(3) *Fluorescence Microscopy: Live-cell fluorescence microscopy could be employed to enhance visualization by labeling FLp53 and the isoforms with different fluorescent markers (e.g., EGFP and mCherry tags).*

We appreciate the suggestion to use live-cell fluorescence microscopy with EGFP and mCherry tags for the visualization FLp53 and its isoforms. While we understand the advantages of live-cell imaging with EGFP / mCherry tags, we restrained us from doing such fusions as the GFP or corresponding protein tags are very big (~30 kDa) with respect to the p53 isoform variants (~30 kDa). Other studies have shown that EGFP and mCherry fusions can alter protein oligomerization, solubility and aggregation(18, 19) Moreover, most fluorescence proteins are prone to dimerization (i.e. EGFP) or form obligate tetramers (DsRed)(20, 21, 22), potentially interfering with the oligomerization and aggregation properties of p53 isoforms, particularly $\Delta 133$ p53 and $\Delta 160$ p53.

Instead, we utilized FLAG- or V5-tag-based immunofluorescence microscopy, a well-established and widely accepted method for visualizing p53 proteins. This method provided precise localization and reliable quantitative data, which we believe meet the needs of the current study. We believe our chosen method is both appropriate and sufficient for addressing the research question.

Reviewer #2 (Significance):

The manuscript by Zhao and colleagues presents a novel and compelling study on the p53 isoforms, $\Delta 133$ p53 and $\Delta 160$ p53, which are associated with aggressive cancer types.

The main objective of the study was to understand how these isoforms exert a dominant negative effect on full-length p53 (FLp53). The authors discovered that the $\Delta 133p53$ and $\Delta 160p53$ proteins exhibit impaired binding to p53-regulated promoters. The data suggest that the predominant mechanism driving the dominant-negative effect is the coaggregation of FLp53 with $\Delta 133p53$ and $\Delta 160p53$.

We sincerely thank the reviewer for the thoughtful and positive comments on our manuscript and for highlighting the significance of our findings on the p53 isoforms, $\Delta 133p53$ and $\Delta 160p53$.

Reviewer #3 (Evidence, reproducibility and clarity):

In this manuscript entitled " $\Delta 133p53$ and $\Delta 160p53$ isoforms of the tumor suppressor protein p53 exert dominant-negative effect primarily by coaggregation", the authors suggest that the $\Delta 133p53$ and $\Delta 160p53$ isoforms have high aggregation propensity and that by co-aggregating with canonical p53 (FLp53), they sequester it away from DNA thus exerting a dominant-negative effect over it.

First, the authors should make it clear throughout the manuscript, including the title, that they are investigating $\Delta 133p53a$ and $\Delta 160p53a$ since there are 3 $\Delta 133p53$ isoforms (α , β , γ), and 3 $\Delta 160p53$ isoforms (α , β , γ).

Thank you for your suggestion. We understand the importance of clearly specifying the isoforms under study. Following your suggestion, we have added α in the title, abstract, and introduction and added the following statement in the Introduction (lines 57-59): "For convenience and simplicity, we have written $\Delta 133p53$ and $\Delta 160p53$ to represent the α isoforms ($\Delta 133p53a$ and $\Delta 160p53a$) throughout this manuscript."

One concern is that the authors only consider and explore $\Delta 133p53a$ and $\Delta 160p53a$ isoforms as exclusively oncogenic and FLp53 dominant-negative while not discussing evidences of different activities. Indeed, other manuscripts have also shown that $\Delta 133p53a$ is non-oncogenic and non-mutagenic, do not antagonize every single FLp53 functions and are sometimes associated with good prognosis. To cite a few examples:

- (1) Hofstetter G. et al. $\Delta 133p53$ is an independent prognostic marker in p53 mutant advanced serous ovarian cancer. Br. J. Cancer 2011, 105, 1593-1599.*
- (2) Bischof, K. et al. Influence of p53 Isoform Expression on Survival in HighGrade Serous Ovarian Cancers. Sci. Rep. 2019, 9, 5244.*
- (3) Knezović F. et al. The role of p53 isoforms' expression and p53 mutation status in renal cell cancer prognosis. Urol. Oncol. 2019, 37, 578.e1578.e10.*
- (4) Gong, L. et al. p53 isoform $\Delta 133p53$ promotes DNA doublestrand break repair to protect cell from death and senescence in response to DNA damage. Cell Res. 2015, 25, 351-369.*
- (5) Gong, L. et al. p53 isoform $\Delta 133p53$ promotes efficiency of induced pluripotent stem cells and ensures genomic integrity during reprogramming. Sci. Rep. 2016, 6, 37281.*
- (6) Horikawa, I. et al. $\Delta 133p53$ represses p53-inducible senescence genes and enhances the generation of human induced pluripotent stem cells. Cell Death Differ. 2017, 24, 1017-1028.*
- (7) Gong, L. p53 coordinates with $\Delta 133p53$ isoform to promote cell survival under low-level oxidative stress. J. Mol. Cell Biol. 2016, 8, 88-90.*

Thank you very much for your comment and for highlighting these important studies.

We agree that $\Delta 133p53$ isoforms exhibit complex biological functions, with both oncogenic and non-oncogenic potentials. However, our mission here was primarily to reveal the molecular mechanism for the dominant-negative effects exerted by the $\Delta 133p53\alpha$ and $\Delta 160p53\alpha$ isoforms on FLp53 for which the $\Delta 133p53\alpha$ and $\Delta 160p53\alpha$ isoforms are suitable model systems. Exploring the oncogenic potential of the isoforms is beyond the scope of the current study and we have not claimed anywhere that we are reporting that. We have carefully revised the manuscript and replaced the respective terms e.g. ‘prooncogenic activity’ with ‘dominant-negative effect’ in relevant places (e.g. line 90). We have now also added a paragraph with suitable references that introduces the oncogenic and non-oncogenic roles of the p53 isoforms.

After reviewing the papers you cited, we are not sure that they reflect on oncogenic /non-oncogenic role of the $\Delta 133p53\alpha$ isoform in different cancer cases. Although our study is not about the oncogenic potential of the isoforms, we have summarized the key findings below:

- (1) Hofstetter et al., 2011: Demonstrated that $\Delta 133p53\alpha$ expression improved recurrence-free and overall survival (in a p53 mutant induced advanced serous ovarian cancer, suggesting a potential protective role in this context.
- (2) Bischof et al., 2019: Found that $\Delta 133p53$ mRNA can improve overall survival in high-grade serous ovarian cancers. However, out of 31 patients, only 5 belong to the TP53 wild-type group, while the others carry TP53 mutations.
- (3) Knezović et al., 2019: Reported downregulation of $\Delta 133p53$ in renal cell carcinoma tissues with wild-type p53 compared to normal adjacent tissue, indicating a potential non-oncogenic role, but not conclusively demonstrating it.
- (4) Gong et al., 2015: Showed that $\Delta 133p53$ antagonizes p53-mediated apoptosis and promotes DNA double-strand break repair by upregulating RAD51, LIG4, and RAD52 independently of FLp53.
- (5) Gong et al., 2016: Demonstrated that overexpression of $\Delta 133p53$ promotes efficiency of cell reprogramming by its anti-apoptotic function and promoting DNA DSB repair. The authors hypothesizes that this mechanism is involved in increasing RAD51 foci formation and decrease γ H2AX foci formation and chromosome aberrations in induced pluripotent stem (iPS) cells, independent of FL p53.
- (6) Horikawa et al., 2017: Indicated that induced pluripotent stem cells derived from fibroblasts that overexpress $\Delta 133p53$ formed noncancerous tumors in mice compared to induced pluripotent stem cells derived from fibroblasts with complete p53 inhibition. Thus, $\Delta 133p53$ overexpression is "non- or less oncogenic and mutagenic" compared to complete p53 inhibition, but it still compromises certain p53-mediated tumor-suppressing pathways. "Overexpressed $\Delta 133p53$ prevented FL-p53 from binding to the regulatory regions of p21WAF1 and miR-34a promoters, providing a mechanistic basis for its dominant-negative inhibition of a subset of p53 target genes."
- (7) Gong, 2016: Suggested that $\Delta 133p53$ promotes cell survival under lowlevel oxidative stress, but its role under different stress conditions remains uncertain.

We have revised the Introduction to provide a more balanced discussion of $\Delta 133p53$'s dule role (lines 62-73):

"The $\Delta 133p53$ isoform exhibit complex biological functions, with both oncogenic and non-oncogenic potentials. Recent studies demonstrate the non-oncogenic yet context-dependent

role of the $\Delta 133p53$ isoform in cancer development. $\Delta 133p53$ expression has been reported to correlate with improved survival in patients with TP53 mutations(23, 24), where it promotes cell survival in a nononcogenic manner(25, 26), especially under low oxidative stress(27). Alternatively, other recent evidences emphasize the notable oncogenic functions of $\Delta 133p53$ as it can inhibit p53-dependent apoptosis by directly interacting with the FLp53 (4, 6). The oncogenic function of the newly identified $\Delta 160p53$ isoform is less known, although it is associated with p53 mutation-driven tumorigenesis(28) and in melanoma cells' aggressiveness(10). Whether or not the $\Delta 160p53$ isoform also impedes FLp53 function in a similar way as $\Delta 133p53$ is an open question. However, these p53 isoforms can certainly compromise p53-mediated tumor suppression by interfering with FLp53 binding to target genes such as p21 and miR-34a(2, 29) by dominant-negative effect, the exact mechanism is not known." On the figures presented in this manuscript, I have three major concerns:

(1) Most results in the manuscript rely on the overexpression of the FLAGtagged or V5-tagged isoforms. The validation of these construct entirely depends on Supplementary figure 3 which the authors claim "rules out the possibility that the FLAG epitope might contribute to this aggregation. However, I am not entirely convinced by that conclusion. Indeed, the ratio between the "regular" isoform and the aggregates is much higher in the FLAG-tagged constructs than in the V5-tagged constructs. We can visualize the aggregates easily in the FLAG-tagged experiment, but the imaging clearly had to be overexposed (given the white coloring demonstrating saturation of the main bands) to visualize them in the V5-tagged experiments. Therefore, I am not convinced that an effect of the FLAG-tag can be ruled out and more convincing data should be added.

Thank you for raising this important concern. We have carefully considered your comments and have made several revisions to clarify and strengthen our conclusions.

First, to address the potential influence of the FLAG and V5 tags on p53 isoform aggregation, we have revised Figure 2 and removed the previous Supplementary Figure 3, where non-specific antibody bindings and higher molecular weight aggregates were not clearly interpretable. In the revised Figure 2, we have removed these potential aggregates, improving the clarity and accuracy of the data.

To further rule out any tag-related artifacts, we conducted a coimmunoprecipitation assay with FLAG-tagged FLp53 and untagged $\Delta 133p53$ and $\Delta 160p53$ isoforms. The results (now shown in the new Supplementary Figure 3) completely agree with our previous result with FLAG-tagged and V5tagged $\Delta 133p53$ and $\Delta 160p53$ isoforms and show interaction between the partners. This indicates that the FLAG / V5-tags do not influence / interfere with the interaction between FLp53 and the isoforms. We have still used FLAGtagged FLp53 as the endogenous p53 was undetectable and the FLAG-tagged FLp53 did not aggregate alone.

In the revised paper, we added the following sentences (Lines 146-152): "To rule out the possibility that the observed interactions between FLp53 and its isoforms $\Delta 133p53$ and $\Delta 160p53$ were artifacts caused by the FLAG and V5 antibody epitope tags, we co-expressed FLAG-tagged FLp53 with untagged $\Delta 133p53$ and $\Delta 160p53$. Immunoprecipitation assays demonstrated that FLAGtagged FLp53 could indeed interact with the untagged $\Delta 133p53$ and $\Delta 160p53$ isoforms (Supplementary Figure 3, lanes 3 and 4), confirming formation of hetero-oligomers between FLp53 and its isoforms. These findings demonstrate that $\Delta 133p53$ and $\Delta 160p53$ can oligomerize with FLp53 and with each other."

Additionally, we performed subcellular fractionation experiments to compare the aggregation and localization of FLAG-tagged FLp53 when co-expressed either with V5-tagged or untagged $\Delta 133p53/\Delta 160p53$. In these experiments, the untagged isoforms also induced FLp53 aggregation, mirroring our previous results with the tagged isoforms (Supplementary Figure 5). We've added this result in the revised manuscript (lines 236-245): "To exclude the

possibility that FLAG or V5 tags contribute to protein aggregation, we also conducted subcellular fractionation of H1299 cells expressing FLAG-tagged FLp53 along with untagged $\Delta 133p53$ or $\Delta 160p53$ at a 1:5 ratio. The results showed (Supplementary Figure 6) a similar distribution of FLp53 across cytoplasmic, nuclear, and insoluble nuclear fractions as in the case of tagged $\Delta 133p53$ or $\Delta 160p53$ (Figure 6A to D). Notably, the aggregation of untagged $\Delta 133p53$ or $\Delta 160p53$ markedly promoted the aggregation of FLAG-tagged FLp53 (Supplementary Figure 6B and D), demonstrating that the antibody epitope tags themselves do not contribute to protein aggregation.”

We’ve also discussed this in the Discussion section (lines 349-356): “In our study, we primarily utilized an overexpression strategy involving FLAG/V5tagged proteins to investigate the effects of p53 isoforms $\Delta 133p53$ and $\Delta 160p53$ on the function of FLp53. To address concerns regarding potential overexpression artifacts, we performed the co-immunoprecipitation (Supplementary Figure 6) and caspase-3 and -7 activity (Figure 7) experiments with untagged $\Delta 133p53$ and $\Delta 160p53$. In both experimental systems, the untagged proteins behaved very similarly to the FLAG/V5 antibody epitopecontaining proteins (Figures 6 and 7 and Supplementary Figure 6). Hence, the C-terminal tagging of FLp53 or its isoforms does not alter the biochemical and physiological functions of these proteins.”

In summary, the revised data set and newly added experiments provide strong evidence that neither the FLAG nor the V5 tag contributes to the observed p53 isoform aggregation.

(2) The authors demonstrate that to visualize the dominant-negative effect, $\Delta 133p53a$ and $\Delta 160p53a$ must be "present in a higher proportion than FLp53 in the tetramer" and the need at least a transfection ratio 1:5 since the 1:1 ration shows no effect. However, in almost every single cell type, FLp53 is far more expressed than the isoforms which make it very unlikely to reach such stoichiometry in physiological conditions and make me wonder if this mechanism naturally occurs at endogenous level. This limitation should be at least discussed.

Thank you for your insightful comment. However, evidence suggests that the expression levels of these isoforms such as $\Delta 133p53$, can be significantly elevated relative to FLp53 in certain physiological conditions(3, 4, 9). For example, in some breast tumors, with $\Delta 133p53$ mRNA is expressed at a much levels than FLp53, suggesting a distinct expression profile of p53 isoforms compared to normal breast tissue(4). Similarly, in non-small cell lung cancer and the A549 lung cancer cell line, the expression level of $\Delta 133p53$ transcript is significantly elevated compared to non-cancerous cells(3). Moreover, in specific cholangiocarcinoma cell lines, the $\Delta 133p53$ /TAp53 expression ratio has been reported to increase to as high as 3:1(9). These observations indicate that the dominant-negative effect of isoform $\Delta 133p53$ on FLp53 can occur under certain pathological conditions where the relative amounts of the FLp53 and the isoforms would largely vary. Since data on the $\Delta 160p53$ isoform are scarce, we infer that the long N-terminal truncated isoforms may share a similar mechanism.

(3) Figure 5C: I am concerned by the subcellular location of the $\Delta 133p53a$ and $\Delta 160p53a$ as they are commonly considered nuclear and not cytoplasmic as shown here, particularly since they retain the 3 nuclear localization sequences like the FLp53 (Bourdon JC et al. 2005; Mondal A et al. 2018; Horikawa I et al, 2017; Joruz S. et al, 2024). However, $\Delta 133p53a$ can form cytoplasmic speckles (Horikawa I et al, 2017) when it colocalizes with autophagy markers for its degradation.

The authors should discuss this issue. Could this discrepancy be due to the high overexpression level of these isoforms? A co-staining with autophagy markers (p62, LC3B) would rule out (or confirm) activation of autophagy due to the overwhelming expression of the isoform.

Thank you for your thoughtful comments. We have thoroughly reviewed all the papers you recommended (Bourdon JC et al., 2005; Mondal A et al., 2018; Horikawa I et al., 2017; Jorruiz S. et al., 2024)(4, 29, 30, 31). Among these, only the study by Bourdon JC et al. (2005) provided data regarding the localization of $\Delta 133p53$ (4). Interestingly, their findings align with our observations, indicating that the protein does not exhibit predominantly nuclear localization in the Figure 8 from Jean-Christophe Bourdon et al. Genes Dev. 2005;19:2122-2137. The discrepancy may be caused by a potentially confusing statement in that paper(4).

The localization of p53 is governed by multiple factors, including its nuclear import and export(32). The isoforms $\Delta 133p53$ and $\Delta 160p53$ contain three nuclear localization sequences (NLS)(4). However, the isoforms $\Delta 133p53$ and $\Delta 160p53$ were potentially trapped in the cytoplasm by aggregation and masking the NLS. This mechanism would prevent nuclear import.

Further, we acknowledge that $\Delta 133p53$ co-aggregates with autophagy substrate p62/SQSTM1 and autophagosome component LC3B in cytoplasm by autophagic degradation during replicative senescence(33). We agree that high overexpression of these aggregation-prone proteins may induce endoplasmic reticulum (ER) stress and activates autophagy(34). This could explain the cytoplasmic localization in our experiments. However, it is also critical to consider that we observed aggregates in both the cytoplasm and the nucleus (Figures 6B and E and Supplementary Figure 6B). While cytoplasmic localization may involve autophagy-related mechanisms, the nuclear aggregates likely arise from intrinsic isoform properties, such as altered protein folding, independent of autophagy. These dual localizations reflect the complex behavior of $\Delta 133p53$ and $\Delta 160p53$ isoforms under our experimental conditions.

In the revised manuscript, we discussed this in Discussion (lines 328-335): “Moreover, the observed cytoplasmic isoform aggregates may reflect autophagy-related degradation, as suggested by the co-localization of $\Delta 133p53$ with autophagy substrate p62/SQSTM1 and autophagosome component LC3B(33). High overexpression of these aggregation-prone proteins could induce endoplasmic reticulum stress and activate autophagy(34). Interestingly, we also observed nuclear aggregation of these isoforms (Figure 6B and E and Supplementary Figure 6B), suggesting that distinct mechanisms, such as intrinsic properties of the isoforms, may govern their localization and behavior within the nucleus. This dual localization underscores the complexity of $\Delta 133p53$ and $\Delta 160p53$ behavior in cellular systems.”

Minor concerns:

- Figure 1A: the initiation of the " $\Delta 140p53$ " is shown instead of " $\Delta 40p53$ "

Thank you! The revised Figure 1A has been created in the revised paper.

- Figure 2A: I would like to see the images cropped a bit higher, so the cut does not happen just above the aggregate bands

Thank you for this suggestion. We've changed the image and the new Figure 2 has been shown in the revised paper.

- Figure 3C: what ratio of FLp53/Delta isoform was used?

We have added the ratio in the figure legend of Figure 3C (lines 845-846) “Relative DNA-binding of the FLp53-FLAG protein to the p53-target gene promoters in the presence of the V5-tagged protein $\Delta 133p53$ or $\Delta 160p53$ at a 1: 1 ratio.”

- Figure 3C suggests that the "dominant-negative" effect is mostly senescencespecific as it does not affect apoptosis target genes, which is consistent with Horikawa et al, 2017 and

Gong et al, 2016 cited above. Furthermore, since these two references and the others from Gong et al. show that $\Delta 133p53a$ increases DNA repair genes, it would be interesting to look at RAD51, RAD52 or Lig4, and maybe also induce stress.

Thank you for your thoughtful comments and suggestions. In Figure 3C, the presence of $\Delta 133p53$ or $\Delta 160p53$ only significantly reduced the binding of FLp53 to the p21 promoter. However, isoforms $\Delta 133p53$ and $\Delta 160p53$ demonstrated a significant loss of DNA-binding activity at all four promoters: p21, MDM2, and apoptosis target genes BAX and PUMA (Figure 3B). This result suggests that $\Delta 133p53$ and $\Delta 160p53$ have the potential to influence FLp53 function due to their ability to form hetero-oligomers with FLp53 or their intrinsic tendency to aggregate. To further investigate this, we increased the isoform to FLp53 ratio in Figure 4, which demonstrate that the isoforms $\Delta 133p53$ and $\Delta 160p53$ exert dominant-negative effects on the function of FLp53.

These results demonstrate that the isoforms can compromise p53-mediated pathways, consistent with Horikawa et al. (2017), which showed that $\Delta 133p53a$ overexpression is "non- or less oncogenic and mutagenic" compared to complete p53 inhibition, but still affects specific tumor-suppressing pathways. Furthermore, as noted by Gong et al. (2016), $\Delta 133p53$'s anti-apoptotic function under certain conditions is independent of FLp53 and unrelated to its dominant-negative effects.

We appreciate your suggestion to investigate DNA repair genes such as RAD51, RAD52, or Lig4, especially under stress conditions. While these targets are intriguing and relevant, we believe that our current investigation of p53 targets in this manuscript sufficiently supports our conclusions regarding the dominant-negative effect. Further exploration of additional p53 target genes, including those involved in DNA repair, will be an important focus of our future studies.

- Figure 5A and B: directly comparing the level of FLp53 expressed in cytoplasm or nucleus to the level of $\Delta 133p53a$ and $\Delta 160p53a$ expressed in cytoplasm or nucleus does not mean much since these are overexpressed proteins and therefore depend on the level of expression. The authors should rather compare the ratio of cytoplasmic/nuclear FLp53 to the ratio of cytoplasmic/nuclear $\Delta 133p53a$ and $\Delta 160p53a$.

Thank you very much for this valuable suggestion. In the revised paper, Figure 5B has been recreated. Changes have been made in lines 214215: "The cytoplasm-to-nucleus ratio of $\Delta 133p53$ and $\Delta 160p53$ was approximately 1.5-fold higher than that of FLp53 (Figure 5B)."

Referees cross-commenting

I agree that the system needs to be improved to be more physiological.

Just to precise, the D133 and D160 isoforms are not truncated mutants, they are naturally occurring isoforms expressed in almost every normal human cell type from an internal promoter within the TP53 gene.

Using overexpression always raises concerns, but in this case, I am even more careful because the isoforms are almost always less expressed than the FLp53, and here they have to push it 5 to 10 times more expressed than the FLp53 to see the effect which make me fear an artifact effect due to the overwhelming overexpression (which even seems to change the normal localization of the protein).

To visualize the endogenous proteins, they will have to change cell line as the H1299 they used are p53 null.

Thank you for these comments. We've addressed the motivation of overexpression in the above responses. We needed to use the plasmid constructs in the p53-null cells to detect the proteins but the expression level was certainly not 'overwhelmingly high'.

First, we tried the A549 cells (p53 wild-type) under DNA damage conditions, but the endogenous p53 protein was undetectable. Second, several studies reported increased $\Delta 133p53$ level compared to wild-type p53 and that it has implications in tumor development(2, 3, 4, 9). Third, the apoptosis activity of H1299 cells overexpressing p53 proteins was analyzed in the revised manuscript (Figure 7). The apoptotic activity induced by FLp53 expression was approximately 2.5 times higher than that of the control vector under identical plasmid DNA transfection conditions (Figure 7). These results rule out the possibility that the plasmid-based expression of p53 and its isoforms introduced artifacts in the results. We've discussed this in the Results section (lines 254269).

Reviewer #3 (Significance):

Overall, the paper is interesting particularly considering the range of techniques used which is the main strength.

The main limitation to me is the lack of contradictory discussion as all argumentation presents $\Delta 133p53a$ and $\Delta 160p53a$ exclusively as oncogenic and strictly FLp53 dominant-negative when, particularly for $\Delta 133p53a$, a quite extensive literature suggests a not so clear-cut activity.

The aggregation mechanism is reported for the first time for $\Delta 133p53a$ and $\Delta 160p53a$, although it was already published for $\Delta 40p53a$, $\Delta 133p53\beta$ or in mutant p53.

This manuscript would be a good basic research addition to the p53 field to provide insight in the mechanism for some activities of some p53 isoforms.

My field of expertise is the p53 isoforms which I have been working on for 11 years in cancer and neuro-degenerative diseases

Thank you very much for your positive and critical comments. We've included a fair discussion on the oncogenic and non-oncogenic function of $\Delta 133p53$ in the Introduction following your suggestion (lines 62-73).

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